

CCNA Exam B

Reorganized to Follow Jeremy's IT Lab CCNA 200-301 Course

74 items - questions, correct answers, and source explanations

What changed: The original Exam B items are reorganized into Jeremy's course order so you can review Boson questions immediately after the matching Jeremy topic. The wording of the source explanations is preserved as closely as possible; obvious PDF text-layer artifacts were normalized for readability.

Lab items: Original questions 1-4 are NetSim lab tasks. The source provides the task prompt but no separate correct-answer/explanation section, so they are reproduced as lab tasks only.

Cross-topic items: A few integrated questions touch multiple subjects. Each appears once under the closest primary Jeremy topic to avoid duplicates.

Selectable/searchable: All rewritten question, answer, and explanation text in this document is real text, not page screenshots.

Course-Order Index

Only Jeremy sections that contain at least one Exam B item are listed. Original Exam B question numbers are retained for traceability.

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Day 2 - Interfaces & Cables

Included Exam B items: Q49, Q63

Original Exam B Question 49

What is the core of a fiber optic cable made of? (Select the best answer.)

- A. cladding
- B. copper
- C. plastic
- D. glass

Correct Answer: D

Explanation

The core of a fiber optic cable is made of glass. The glass core is surrounded by a cladding made of lower-refractive glass or plastic. The cladding is surrounded by a polymer coating, strengthening fibers, and a cable jacket. The core carries data by using light pulses transmitted by a laser or a light-emitting diode (LED). The cladding reflects the light back into the core so that the signal can be carried over long distances. Because fiber cables transmit data by light, they are not susceptible to electromagnetic interference (EMI).

The core of a fiber optic cable is not made of cladding, plastic, or copper. The core of a coaxial cable is made of copper. Coaxial cables are flexible plastic cables that consist of an inner copper core surrounded by a bedding of a dielectric insulator. The insulator is covered with a plastic sheath or a woven copper shield. Although coaxial cable was used in older computer networks, it is now most commonly used for carrying cable television signals. Coaxial cable is susceptible to EMI.

Ethernet cables typically transmit data over eight copper wires that are twisted together in pairs.

Unshielded twisted pair (UTP) cables are susceptible to EMI; shielded twisted pair (STP) cables are less susceptible to EMI because they contain an extra layer of shielding, which is typically made of aluminum foil.

Original Exam B Question 63

You are connecting two Catalyst 9500 switches with fiber-optic cable. When you boot SwitchA, you receive a SYS-3-TRANSCEIVER_NOTAPPROVED error.

Which of the following is most likely the cause of the problem? (Select the best answer.)

- A. You have installed a third-party SFP module.
- B. You have installed the SFP module upside down.
- C. There is a physical problem with the fiber cable.
- D. You have connected a cable to an incorrect port.

Correct Answer: A

Explanation

You have most likely installed a third-party Small Form-Factor Pluggable (SFP) transceiver module in SwitchA if you receive a SYS-3-TRANSCEIVER_NOTAPPROVED error when you boot SwitchA. Cisco devices do not officially support the use of third-party SFP modules. An SFP module is a hot-pluggable device that enables a switch, router, or other device to accept connections from different types of media, such as single-mode fiber (SMF), multimode fiber (MMF), or twisted-pair Ethernet cables.

An SFP module that is installed in a Cisco device stores identifying information, such as the module serial number, vendor name, and security code. When a switch detects the insertion of an SFP module, the switch software attempts to read the identifying information stored on the SFP module. If the information is not valid or not present, the switch software will report the SYS-3-TRANSCEIVER_NOTAPPROVED error.

You can issue the show interfaces command to identify the type of SFP module that is installed in a particular interface. For example, the underlined text in the following sample output indicates that a Cisco SFP-10G-LR module is currently installed in the TenGigabitEthernet 1/0/10 interface:

```
SwitchA#show interfaces TenGigabitEthernet 1/0/10
```

```
TenGigabitEthernet1/0/10 is up, line protocol is up (connected)
```

```
Hardware is Ten Gigabit Ethernet, address is 00a5.bf9c.2aa3 (bia 00a5.bf9c.2aa3)
```

MTU 1500 bytes, BW 10000000 Kbit/sec, DLY 10 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive not set
Full-duplex, 10Gb/s, link type is auto, media type is SFP-10GBase-LR
input flow-control is on, output flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:13, output 00:00:00, output hang never
Last clearing of "show interface" counters never

It is important to ensure that compatible SFP modules are installed at each end of a link. For example, the SFP-10Gbase-LR module in the sample output above requires the use of single-mode cable and a compatible SFP on the far end of the link. However, if an SFP that required multimode cable was installed on the far end of the link, there would likely be intermittent connectivity issues if the link was somehow forced into an active state.

The switch would not report a SYS-3-TRANSCEIVER_NOTAPPROVED error if you had connected a cable to an incorrect port. If you connected a cable to the wrong SFP module port, you would most likely notice that the ports on the switches are up, but the line protocol is down.

The switch would not report a SYS-3-TRANSCEIVER_NOTAPPROVED error if there were a physical problem with the fiber cable. If the fiber cable were broken, you would notice that the port status light-emitting diodes (LEDs) on the SFP modules are not lit.

The switch would not report a SYS-3-TRANSCEIVER_NOTAPPROVED error if you had installed the SFP module upside down. Instead, the switch would not recognize the SFP module, and the output from show commands would contain no information about the module.

Days 5-6 - Ethernet LAN Switching

Included Exam B items: Q7

Original Exam B Question 7

Which of the following statements are true regarding switches? (Select 2 choices.)

- A. Switches forward broadcast frames and multicast frames.
- B. Switches segment a network into separate broadcast domains.
- C. Switches help reduce collisions on a network.
- D. Switches operate at the Network layer of the OSI model.
- E. Switches rely on the destination IP address in order to make forwarding decisions.

Correct Answer: A, C

Explanation

Switches can help reduce collisions on a network and can forward broadcast and multicast frames. Switches typically operate at Open Systems Interconnection (OSI) Layer 2, or the Data Link layer, not at the Network layer, although some Cisco switches can be configured to function as Network layer devices. As Layer 2 devices, switches make forwarding decisions based on the Media Access Control (MAC) address of the destination device. When a switch receives a frame, the switch adds the source MAC address to the switching table, if the address is not already in the table. This table entry tells the switch to which port to send frames that are destined for that address. Then the switch will check the switching table to see whether the destination MAC address is listed. If it is, the switch will direct the frame to the appropriate port. If the destination address is not listed, the switch will broadcast the frame out all ports except the port from which the frame was received. Switches can forward both broadcast and multicast frames, although many Cisco switches can be configured so that they do not forward all multicast frames. Ethernet frames use unicast addresses to send data to a single device, and multicast addresses are used to simultaneously send the same data to a group of devices, thereby conserving bandwidth.

Switches, as well as bridges, divide hosts into separate collision domains, although a single broadcast domain exists for the hosts that are connected to a switch or bridge. However, virtual local area networks (VLANs) can be configured on switches to divide the hosts into separate broadcast domains, whereas VLANs cannot be configured on bridges. Switches also support microsegmentation, a technique by which a separate, dedicated network segment is created for each user.

Routers are Layer 3, or Network layer, devices that are responsible for path determination and packet switching. Routers use the Internet Protocol (IP) address of the destination to make forwarding decisions about the packets and then create a separate broadcast domain for each interface.

Broadcast packets from one interface are not forwarded out the other interfaces. Routers also perform packet switching by forwarding packets from one network to another network. Routers can use routing protocols, such as Open Shortest Path First (OSPF) and Routing Information Protocol (RIP), to help determine path selection for received packets.

Days 7-8 - IPv4 Addressing

Included Exam B items: Q12

Original Exam B Question 12

Which of the following IP address blocks are defined by RFC 1918 to conserve globally unique IP addresses?

(Select 2 choices.)

- A. 198.51.100.0/24
- B. 192.168.0.0/16
- C. 192.0.2.0/24
- D. 172.16.0.0/12
- E. 203.0.113.0/24

Correct Answer: B, D

Explanation

The 172.16.0.0/12 and 192.168.0.0/16 Internet Protocol (IP) address blocks are defined by Request For Comments (RFC) 1918 to conserve globally unique IP addresses. The Internet Assigned Numbers Authority (IANA) defines three IP address blocks for use in private networks. Because these IP address blocks are intended for private local area network (LAN) use, they are commonly filtered by edge routers so that packets with private IP addresses are not inadvertently routed across the Internet. The following list indicates the IP address blocks that are reserved for private, internal use:

- 10.0.0.0/8
- 172.16.0.0/12
- 192.168.0.0/16

The 192.0.2.0/24, 198.51.100.0/24, and 203.0.113.0/24 IP address blocks are reserved by IANA for documentation and example code. In order to mitigate the use of real or private IP addresses in documentation, IANA has defined these three IP address blocks for use in documentation and example code. In addition, IANA has reserved the domain names example.com and example.net for use in documentation and example code. The three reserved address blocks are described in RFC 3757, and their reservation is specified in RFC 5735.

Days 11-12 - Routing Fundamentals, Static Routing & Life of a Packet

Included Exam B items: Q5, Q30, Q73

Original Exam B Question 5

You issue the show ip route command on RouterA and receive the following partial output:

S 10.20.0.0/22 [1/0] via 192.168.10.2

R 10.20.0.0/24 [1280/3] via 192.168.10.3, 00:33:38, Serial0/3

D 10.20.0.0/26 [90/2809856] via 192.168.10.4, 00:02:14, Serial0/4

O 10.20.0.0/28 [110/64] via 192.168.10.1, 00:02:38,

RouterA receives a packet that is destined for 10.20.0.14.

Which of the following routes will RouterA use to send the packet? (Select the best answer.)

- A. the OSPF route, because it is the route with the longest prefix match
- B. the RIP route, because it has the highest AD
- C. the static route, because static routes are preferred over dynamic routes
- D. the EIGRP route, because it has the lowest AD

Correct Answer: A

Explanation

RouterA will use the Open Shortest Path First (OSPF) route, because it is the route with the longest prefix match. When a packet is sent to a router, the router checks the routing table to see whether the next-hop address for the destination network is known. If multiple routes to a destination are known, the most specific route is used. Therefore, the following rules apply on RouterA:

- Packets sent to the 10.20.0.0/28 network use the OSPF route. This includes destination addresses from 10.20.0.0 through 10.20.0.15.

- Packets sent to the 10.20.0.0/26 network, except those sent to the 10.20.0.0/28 network, use the Enhanced Interior Gateway Routing Protocol (EIGRP) route. This includes destination addresses from 10.20.0.16 through 10.20.0.63.

- Packets sent to the 10.20.0.0/24 network, except those sent to the 10.20.0.0/26 network, use the Routing Information Protocol (RIP) route. This includes destination addresses from 10.20.0.64 through 10.20.0.255.

- Packets sent to the 10.20.0.0/22 network, except those sent to the 10.20.0.0/24 network, use the static route. This includes destination addresses from 10.20.1.0 through 10.20.3.255.

- Packets sent to any destination not listed in the routing table are forwarded to the default gateway, if one is configured.

Because the most specific route to 10.20.0.14 is the route toward the 10.20.0.0/28 network, RouterA will forward a packet destined for 10.20.0.14 to the Serial0/1 interface.

RouterA will not use the EIGRP route to send a packet that is destined for 10.20.0.14. Administrative distance (AD) values are used only to determine which route is placed in the routing table when multiple routes to a destination are known. However, a router considers routes with different prefix lengths as separate routes. If OSPF, EIGRP, and RIP had each advertised routes to 10.20.0.0/28, the EIGRP route would have been selected because EIGRP has the lowest AD. The following list contains the most commonly used ADs:

Route Source AD

Directly connected route 0

Static route 1

EIGRP summary route 5

eBGP 20

Internal EIGRP 90

IGRP 100

OSPF 110

IS-IS 115

RIP 120

External EIGRP 170

iBGP 200

Unknown 255

RouterA will not use the RIP route to send a packet that is destined for 10.20.0.14. Routes with longer

prefix lengths are preferred over routes with shorter prefix lengths, and routes with lower ADs are preferred over routes with higher ADs.

RouterA will not use the static route to send a packet that is destined for 10.20.0.14. If the static route were configured so that the destination network were 10.20.0.0/28, the static route would be preferred over the OSPF route because static routes have a lower AD than dynamic routes.

Original Exam B Question 30

IP routing is not enabled on a Cisco router that you administer. You want to create a gateway of last resort on the router.

Which of the following commands should you issue to accomplish your goal? (Select the best answer.)

- A. `ip route 0.0.0.0 0.0.0.0 fa/0`
- B. `ip default-gateway 10.10.100.1`
- C. `ip default-network 10.10.100.0`
- D. `ip route 0.0.0.0 0.0.0.0 10.10.100.1`

Correct Answer: B

Explanation

You should issue the `ip default-gateway 10.10.100.1` command to create a gateway of last resort on the router in this scenario. The gateway of last resort is used when no route can be identified for routing packets. If a route cannot be determined and the gateway of last resort has been defined, packets are routed to the gateway's address. The `ip default-gateway` command should be used to configure a gateway of last resort when Internet Protocol (IP) routing is disabled on a Cisco router or switch.

You should not issue the `ip default-network 10.10.100.0` command to create a gateway of last resort on the router in this scenario, because the `ip default-network` command can be used only on devices that have IP routing enabled. Because IP routing is disabled on the router in this scenario, you cannot use the `ip default-network` command to create a gateway of last resort on the router. The `ip default-network` command is issued to flag routes in the routing table as candidates for the default route. If the `ip default-network` command is issued with a network address that matches an entry appearing in the routing table, that route becomes the default route. If there is no matching entry in the routing table, the route will not become a default route. You can issue multiple `ip default-network` commands on a router; the router will use administrative distances and metrics to determine the best default route.

You should not issue the `ip route` command to create a gateway of last resort on the router in this scenario, because the `ip route` command can be used only on devices that have IP routing enabled. You can create a static gateway of last resort on a router by issuing the `ip route` command with a static route followed by an IP address. For example, you could issue the `ip route 0.0.0.0 0.0.0.0 10.10.100.1` command to create a static gateway of last resort to the next-hop router at 10.10.100.1. Alternatively, you could specify a physical interface on the router as the gateway. For example, you could issue the `ip route 0.0.0.0 0.0.0.0 fa0/0` command to specify the Fast Ethernet 0/0 interface on the local router as the gateway of last resort; packets with destinations not found in the routing table will be forwarded to the FastEthernet 0/0 interface.

Original Exam B Question 73

RouterA receives routes to the following overlapping networks:

- 192.168.1.0/24
- 192.168.1.0/25
- 192.168.1.0/26
- 192.168.1.0/28

Each of the routes is received from a different routing protocol.

Which of the following routes will RouterA install in the routing table? (Select the best answer.)

- A. all of the routes
- B. the route with the longest prefix match
- C. the route with the lowest AD
- D. the route with the highest AD
- E. the route with the shortest prefix match

Correct Answer: A

Explanation

RouterA will install all of the routes in the routing table. When multiple routes to overlapping networks exist, a router will prefer the most specific route, which is the route with the longest prefix match. For example, if RouterA receives a packet to 192.168.1.4, it will send the packet to the 192.168.1.0/28 route; if RouterA receives a packet to 192.168.1.20, it will send the packet to the 192.168.1.0/26 route. RouterA will not install only the route with the longest or shortest prefix match. RouterA will not install only the route with the highest or lowest administrative distance (AD), because the routes target separate destination networks. When multiple routes to the same destination network exist and each route uses a different routing protocol, a router will install only the route from the routing protocol with the lowest AD. The following list contains the most commonly used ADs:

Route Source AD

Directly connected route 0

Static route

EIGRP summa

eBGP

Internal EIGRP

IGRP.

IS-IS

RIP

External EIGRP

iBGP

Unknown 255

ADs for a routing protocol can be manually configured by issuing the distance command in router configuration mode. For example, to change the AD of Open Shortest Path First (OSPF) process 1 from 110 to 80, you should issue the following commands:

```
RouterA(config)#router ospf 1
```

```
RouterA(config-router)#distance 80
```

Days 13-15 - Subnetting

Included Exam B items: Q11, Q42, Q48, Q52

Original Exam B Question 11

Given a host IP address of 48.25.24.71/21, what is the last available host address in this network? (Select the best answer.)

- A. 48.25.24.240
- B. 48.25.24.255
- C. 48.25.25.255
- D. 48.25.31.254
- OE 48.25.25.254
- F. 48.25.24.127

Correct Answer: D

Explanation

The last available address for the subnetwork containing the host 48.25.24.71/21 is 48.25.31.254. An Internet Protocol (IP) address consists of four groups of 8 binary bits, or 32 bits total. Each bit contains either a 1 or a 0 binary value. The IP address consists of two parts, a network portion and a host portion, which are divided by the use of a subnet mask. Like the IP address, the subnet mask also has four groups of 8 binary bits, each bit corresponding to either a 1 or a 0, for a total of 32 bits. Because each group contains 8 bits of information, the groups are referred to as octets and range from 0 through 255 in decimal value.

To determine the subnetwork address range of a given IP address/subnet mask combination, you must first identify the interesting octet within the subnet mask. The interesting octet is the first octet that contains a decimal value other than 255 or 0.

The subnet mask in this scenario, /21, is presented in Classless Inter-Domain Routing (CIDR) notation. The subnet mask will need to be converted from CIDR notation to dotted decimal notation in order for you to calculate networks and hosts.

To convert /21 from CIDR notation to dotted decimal notation, begin at the left and set the first 21 bits to a value of 1. These bits identify the network portion of the IP address. The remaining 11 bits will be set to 0.

/21 = 11111111.11111111.1111000.00000000

Binary bit weight increases in significance from right to left, with the leftmost bit in each octet worth a decimal value of 128 and the rightmost bit worth a decimal value of 1. The decimal value for each octet is computed by adding up the bit weight for any bit containing a 1 within the octet. The following exhibit displays how to calculate the decimal value of the subnet mask octets based on the binary value assigned to each bit:

BIT WEIGHT DECIMAL

ce es ae es

EE EEEEEEEEEEEEEEE

KE EEEEEEEE EE EE

EEE ERR

oo

/21 = 11111111.11111111.1111000.00000000 = 255.255.248.0

Now that the subnet mask is in dotted decimal notation, the interesting octet is easily identified, because the third octet has a decimal value of 248, not 255 or 0.

Once the interesting octet has been identified, the network numbers are determined by the weight of the least significant bit, or the rightmost bit, in the interesting octet that is set to 1. Another way to determine the network numbers is to simply subtract the decimal value of the interesting octet from 256. The difference between 256 and 248 is 8; therefore, the networks will be arranged in multiples of 8.

En En EEEB

A partial list of the available networks using the 255.255.248.0 subnet mask in this scenario contains the following:

48.25.0.0

48.25.8.0

48.25.16.0

48.25.24.0

48.25.32.0

..and so on

The total number of hosts can be determined by the number of bits equal to 0 in the binary subnet mask. You can then calculate the number of hosts for a given subnetwork by using the formula $2^n - 2$, where n is the number of bits equal to 0 in the subnet mask.

/21 = 11111111.11111111.1111000.00000000

There are 11 bits equal to 0 in the /21 subnet mask. Using the $2^n - 2$ formula, $2,048 - 2 = 2,046$.

Therefore, 2,046 hosts are available for each subnetwork when a subnet mask of /21 is applied. You must subtract 2 from the number of available hosts, because the first address is the subnetwork address and the last address is the broadcast address.

Likewise, you can determine the number of subnetworks that are available by examining the network mask. Because you are working with a Class A network, the first octet is excluded from the calculation.

The formula to calculate the number of networks is 2^n , where n is the number of bits equal to 1.

XXXXXXXX.11111111.1111000.00000000

Excluding the first octet, 13 bits that equal 1 are still in the network mask. Using the formula 2^n , you should find that 8,192 networks are available when a subnet mask of 255.255.248.0 is applied to a Class A network.

Applying a subnet mask larger than the default subnet mask for a given network class is known as supernetting. This technique allows network engineers and administrators to combine and divide networks without regard to classful network boundaries, thus making more efficient use of IP address space.

Based on the information above, an IP address of 48.25.24.71/21 is located in the 48.25.24.0 network.

The first address in the subnetwork is 48.25.24.1, and the last usable host address is 48.25.31.254.

Original Exam B Question 42

Select the subnet masks, and place them over the number of host addresses that the subnet mask can support. Not all subnet masks will be used.

Correct Answer: /25 - 126 host addresses

/23 - 510 host addresses

/22 - 1,022 host addresses

/20 - 4,094 host addresses

/19 - 8,190 host addresses

Explanation

A subnet mask specifies how many bits belong to the network portion of a 32-bit Internet Protocol version 4 (IPv4) address. The remaining bits in the Internet Protocol (IP) address belong to the host portion of the IP address. To determine how many host addresses are defined by a subnet mask, use the formula $2^n - 2$, where n is the number of bits in the host portion of the address.

A /19 subnet mask uses 13 bits for host addresses. Therefore, $2^{13} - 2$ equals 8,190 valid host addresses.

A /20 subnet mask uses 12 bits for host addresses. Therefore, $2^{12} - 2$ equals 4,094 valid host addresses.

A /22 subnet mask uses 10 bits for host addresses. Therefore, $2^{10} - 2$ equals 1,022 valid host addresses.

A /23 subnet mask uses 9 bits for host addresses. Therefore, $2^9 - 2$ equals 510 valid host addresses.

A /25 subnet mask uses 7 bits for host addresses. Therefore, $2^7 - 2$ equals 126 valid host addresses.

Although it is important to learn the formula for calculating valid host addresses, the following list demonstrates the relationship between subnet masks and valid host addresses:

- /16 subnet mask contains $2^{16} - 2 = 65,534$ host addresses
- /17 subnet mask contains $2^{15} - 2 = 32,766$ host addresses
- /18 subnet mask contains $2^{14} - 2 = 16,382$ host addresses
- /19 subnet mask contains $2^{13} - 2 = 8,190$ host addresses
- /20 subnet mask contains $2^{12} - 2 = 4,094$ host addresses
- /21 subnet mask contains $2^{11} - 2 = 2,046$ host addresses
- /22 subnet mask contains $2^{10} - 2 = 1,022$ host addresses
- /23 subnet mask contains $2^9 - 2 = 510$ host addresses

e /24 subnet mask contains $2^8 - 2 = 254$ host addresses
 e /25 subnet mask contains $2^7 - 2 = 126$ host addresses
 • /26 subnet mask contains $2^6 - 2 = 62$ host addresses
 e /27 subnet mask contains $2^5 - 2 = 30$ host addresses
 • /28 subnet mask contains $2^4 - 2 = 14$ host addresses
 • /29 subnet mask contains $2^3 - 2 = 6$ host addresses
 e /30 subnet mask contains $2^2 - 2 = 2$ host addresses

Original Exam B Question 48

Which of the following pairs of IP addresses could you use at each end of a point-to-point link? (Select the best answer.)

- A. 10.1.1.127/30 and 10.1.1.128/30
- B. 10.1.1.153/30 and 10.1.1.154/30
- O 10.1.1.236/30 and 10.1.1.237/30
- D. 10.1.1.2/30 and 10.1.1.3/30
- E. 10.1.1.62/30 and 10.1.1.65/30

Correct Answer: B

Explanation

You could use the 10.1.1.153/30 and 10.1.1.154/30 Internet Protocol (IP) addresses for each end of a point-to-point link. A /30 subnet mask is typically used for point-to-point links so that the available IP address space is conserved. Networks that are subnetted by using /30 masks are separated into groups of four addresses each. The first address is the subnet address, the next two addresses are valid host addresses, and the last address is the broadcast address for the subnet. Because there are four addresses in each subnet, each subnet address is divisible by four.

The following list indicates how many IP addresses exist within each subnet, based on the subnet mask:

e /24 subnet mask: 256 addresses in each subnet
 /25 subnet mask: 128 addresses in each subnet
 /26 subnet mask: 64 addresses in each subnet
 /27 subnet mask: 32 addresses in each subnet
 /28 subnet mask: 16 addresses in each subnet
 /29 subnet mask: 8 addresses in each subnet
 /30 subnet mask: 4 addresses in each subnet

To determine the number of valid host addresses for each of the subnets in the list, subtract 2 from the total number of addresses: 1 for the subnet address and 1 for the broadcast address.

The 10.1.1.153/30 and 10.1.1.154/30 IP addresses can be used as host addresses for each end of a point-to-point link for the following reasons. First, neither address is a subnet address or a broadcast address; second, both host addresses are on the same subnet. The subnet address is 10.1.1.152/30; the last octet, 152, is divisible by 4. The broadcast address for the subnet is 10.1.1.155/30. The subnet address for the next available /30 subnet is 10.1.1.156/30.

The 10.1.1.3/30 IP address cannot be used as a host address, because it is a broadcast address. The subnet address is 10.1.1.0/30, the valid host addresses are 10.1.1.1/30 and 10.1.1.2/30, and the broadcast address is 10.1.1.3/30.

The 10.1.1.62/30 and 10.1.1.65/30 IP addresses are valid host addresses. However, these two addresses cannot be used for each end of a point-to-point link, because they reside within different subnets. The 10.1.1.62/30 address resides within the 10.1.1.60/30 subnet, and the 10.1.1.65/30 address resides within the 10.1.1.64/30 subnet.

The 10.1.1.127/30 and 10.1.1.128/30 IP addresses cannot be used as host addresses. The 10.1.1.127/30 address is the broadcast address for the 10.1.1.124/30 subnet, and the 10.1.1.128/30 address is the subnet address for the 10.1.1.128/30 subnet.

The IP address 10.1.1.236/30 cannot be used as a host address, because it is the subnet address for the 10.1.1.236/30 subnet. The last octet, 236, is divisible by 4; therefore, the address 10.1.1.236/30 is the subnet address.

Original Exam B Question 52

Which of the following dotted decimal subnet masks is equivalent to the binary subnet mask 11111111.11111111.11110000.00000000? (Select the best answer.)

- A. 255.255.240.0
- B. 255.255.224.0
- C. 255.255.248.0
- D. 255.255.192.0

Correct Answer: A

Explanation

The subnet mask 255.255.240.0 is equivalent to 11111111.11111111.11110000.00000000. There are three ways to express a subnet mask: binary notation, dotted decimal notation, and Classless Inter-Domain Routing (CIDR) notation. Binary notation uses 1s and 0s. Dotted decimal notation uses decimal numbers from 0 through 255. CIDR notation is a shortcut notation that indicates how many bits in the decimal subnet mask are set to a value of 1.

When converting binary notation to dotted decimal notation, you must convert each eight-digit set, or octet, individually. To convert a binary number to decimal, you should calculate the value of each of the binary digits and add them together. The first digit in an eight-digit binary number is worth 128 in decimal, the second digit is worth 64 in decimal, and so on. For example, the binary number 11111111 is calculated by the following method:

128 (1 x 128)
+64 (1 x 64)
+32 (1 x 32)
+16 (1 x 16)
+8 (1x 8)
+4 (1x 4)
+2 (1 x 2)
+1(1x 1)
255

Knowing the decimal value of each of the binary digits is the key to converting from binary to decimal. If you see a 1 for a binary place value, you should add the corresponding decimal equivalent for that place value. If you see a 0 for a binary place value, you should not add anything.

To convert the binary number 11110000, use the following calculation:

128 (1 x 128)
+64 (1 x 64)
+32 (1 x 32)
+16 (1 x 16)
+ 0 (0 x 8)
+0 (0x 4)
+ 0 (0 x 2)
+0(0x 1)
240

The binary number 00000000 is equal to the decimal number 0. Therefore, the binary number 11111111.11111111.11110000.00000000 is equal to the dotted decimal number 255.255.240.0.

A subnet mask in CIDR notation begins with a forward slash and a number that indicates how many digits of a subnet mask are used to specify the network portion of an IP address. When calculating the CIDR notation for a subnet mask, you must determine how many binary digits are set to a value of 1. For example, the binary number 11111111.11111111.11110000.00000000 contains 20 consecutive binary 1s; therefore, the subnet mask in CIDR notation is /20.

The dotted decimal subnet mask 255.255.192.0 is equivalent to 11111111.11111111.11000000.00000000. You would use the following calculation to convert the third octet 11000000 to decimal:

128 (1 x 128)
+64 (1 x 64)
192

The dotted decimal subnet mask 255.255.224.0 is equivalent to 11111111.11111111.11100000.00000000. You would use the following calculation to convert the third octet 11100000 to decimal:

128 (1 x 128)
+64 (1 x 64)
+32 (1 x 32)

224

The dotted decimal subnet mask 255.255.248.0 is equivalent to

11111111.11111111.11111000.00000000. You would use the following calculation to convert the third octet 11111000 to decimal:

128 (1 x 128)

+64 (1 x 64)

+32 (1 x 32)

+16 (1 x 16)

+8 (1x 8)

248

Days 16-18 - VLANs

Included Exam B items: Q4, Q31

Original Exam B Question 4

Your company wants to add data and voice VLANs to SwitchA and SwitchB.

According to the following table, you must create and name each VLAN, configure the appropriate ports as access ports, and assign the VLANs to the corresponding ports:

·VLAN |Name Ports ·SwitchB Ports

3 Development Fa0/1-2 Fa0/2

4 Manufacturing Fa0/4

5 Finance Fa0/3 Fa0/1, Fa0/3

6 HR Fa0/4 Fa0/5

A trunk link has already been established between the FastEthernet 0/0 interfaces of SwitchA and SwitchB. However, you should configure the trunk so that it accepts traffic for only these newly created VLANs as well as the native VLAN.

Guidelines | Topology | Tasks ·SwitchA |

Source result: NetSim lab task - no separate answer/explanation is included in the uploaded Exam B PDF.

Original Exam B Question 31

You have issued the show running-config command on SwitchA and have received the following partial output:

```
interface FastEthernet0/8
```

```
switchport trunk encapsulation dot1q
```

```
switchport access vlan 26
```

```
spanning-tree portfast
```

Why would port FastEthernet 0/8 not be displayed in the output of the show vlan command on SwitchA?

(Select the best answer.)

A. FastEthernet 0/8 is configured as a trunk port.

B. FastEthernet 0/8 is not configured with an IP address.

C. FastEthernet 0/8 is configured to use PortFast.

D. FastEthernet 0/8 is configured as an access port.

Correct Answer: A

Explanation

Port FastEthernet 0/8 would not be displayed in the output of the show vlan command on SwitchA because FastEthernet 0/8 is configured as a trunk port. Only access ports are displayed in the output of the show vlan command, as shown in the following output:

```
SwitchA#show vlan
```

```
VLAN Name Status Ports
```

```
a default active Fa0/2, Fa0/4, Fa0/5, Fae/6
```

```
Fa0/7, Fa0/9, Fa0/10,
```

```
Fa0/12, Gie/1, Gie/2
```

```
2 VLAN0GO2 active
```

```
10 VLAN001e active Fae/3
```

```
28 VLAN0e82e active
```

```
1002 fddi-default active
```

```
1003 token-ring-default active
```

```
1004 fddinet-default active
```

```
1005 trnet-default active
```

```
VLAN Type SAID MTU Parent RingNo BridgeNo Stp BrdgMode Trans1 Trans2
```

```
10 enet 100010 1500 - - - -
```

```
28 enet 100020 1500 - - - -
```

```
1002 fddi 101002 1566 - - - -
```

```
10e03 tr 101003 1500 - - - -
```

```
1004 fdnet 101004 1506 - - - ieee -
```

```
1005 trnet 101005 1566 - - - ibm -
```

To display information about trunk ports, you should issue the show interfaces trunk command, as shown in the following output:

```
SwitchA#show interfaces trunk
Port Mode Encapsulation Status Native vlan
Fa0/1 on 802.1q trunking 1
Fa0/8 on 802.1q trunking 1
Port Vlans allowed on trunk
Fa0/1 1-4094
Fa0/8
Port Vlans allowed and active in management domain
Fa0/1 1,2,10,20
Fa0/8 1,2,10,20
Port Vlans in spanning tree forwarding state and not pruned
Fa0/1 1,2,10,20
Fa0/8 1,2,10,20
```

Even though FastEthernet 0/8 is configured with the switchport access vlan 20 command, which is typically issued on access ports, FastEthernet 0/8 is also configured with the switchport mode trunk command, which configures FastEthernet 0/8 to be a trunk port, not an access port. To configure FastEthernet 0/8 to be an access port, you should issue the switchport mode access command from interface configuration mode.

Although port FastEthernet 0/8 is not configured with an Internet Protocol (IP) address, an IP address is not required for a switch port to be displayed by the show vlan command. Access ports operate at Layer 2; IP addresses are Layer 3 addresses.

Although port FastEthernet 0/8 is configured to use PortFast, PortFast will not prevent a switch port from being displayed by the show vlan command. PortFast is used to immediately place a switch port into a forwarding state, skipping the Spanning Tree Protocol (STP) listening and learning states. PortFast should be used only on ports that are connected to an end host, such as an IP phone, a client workstation, or a server. Conversely, PortFast should not be enabled on a port that is connected to a switch or other networking device. If you enable PortFast on such a port, you risk creating bridging loops because the port is permanently in the STP forwarding state.

Days 20-21 - Spanning Tree Protocol (STP)

Included Exam B items: Q19, Q53, Q55

Original Exam B Question 19

You connect a new, unconfigured switch to an existing switch's FastEthernet 0/1 interface. That interface was previously connected to an end user's workstation. You notice that the FastEthernet 0/1 interface on the existing switch enters the err-disabled state.

Which of the following are the most likely causes of the problem? (Select 2 choices.)

- A. PortFast is enabled on the FastEthernet 0/1 interface.
- B. The interface on the new switch is a statically configured trunk port.
- C. Loop guard is enabled on the FastEthernet 0/1 interface.
- D. Root guard is enabled on the FastEthernet 0/1 interface.
- E. BPDU guard is enabled on the FastEthernet 0/1 interface.

Correct Answer: A, E

Explanation

Of the available choices, the most likely cause of the problem is that the existing switch's FastEthernet 0/1 interface has PortFast and BPDU guard enabled. BPDU guard is used to disable ports that erroneously receive bridge protocol data units (BPDUs). BPDU guard is typically applied to edge ports that have PortFast enabled. Because PortFast automatically places ports into a forwarding state, a switch that has been connected to a PortFast-enabled port could cause switching loops. However, when BPDU guard is applied, the receipt of a BPDU on a port will result in the port being placed into the err-disabled state, which prevents loops from occurring. When such a port receives a BPDU, BPDU guard immediately puts that port into the err-disabled state and shuts down the port. The port must then be manually re-enabled, or it can be recovered automatically by configuring the errdisable recovery cause bpduguard command and the errdisable recovery interval interval command. BPDU guard should be enabled on ports that have been enabled with PortFast so that BPDU guard can prevent a rogue switch from modifying the Spanning Tree Protocol (STP) topology. PortFast is a feature that provides immediate accessibility to the network for edge ports, such as access ports that are connected to end-user workstations. PortFast transitions the port into the STP forwarding state without going through the STP listening and learning states.

It is not likely that the interface on the new switch is a statically configured trunk port. Depending on the Cisco hardware model, it is possible that Dynamic Trunking Protocol (DTP) is by default configured in dynamic desirable mode, which means that the new device would automatically send out DTP frames in an attempt to negotiate trunking mode. If the FastEthernet 0/1 interface on the existing switch is not statically configured as an access port with the switchport mode access command, the ports could negotiate a trunk link.

It is not likely that loop guard is enabled on the FastEthernet 0/1 interface. The loop guard feature prevents nondesignated ports from inadvertently forming bridging loops if the steady flow of BPDUs is interrupted. When the port stops receiving BPDUs, loop guard puts the port into the loop-inconsistent state, which keeps the port in a blocking state. After the port starts receiving BPDUs again, loop guard automatically re-enables the port so that it transitions through the normal STP states. You can enable loop guard for the entire switch by issuing the spanning-tree loopguard default command in global configuration mode, or you can enable loop guard for specific ports by issuing the spanning-tree guard loop command in interface configuration mode.

It is not likely that root guard is enabled on the FastEthernet 0/1 interface. Root guard is used to prevent newly introduced switches from being elected as the new root switch. This allows administrators to maintain control over which switch is the root. When STP is used, the device with the lowest switch priority is elected the root. If a new device is added to the network with a lower priority than the current root, it will become the new root. However, this could cause the network to reconfigure in unintended ways. To prevent this, root guard can be applied. Root guard is applied on a per-port basis by issuing the spanning-tree guard root command. If root guard is enabled on a loop guard-enabled port, loop guard will be automatically disabled.

Original Exam B Question 53

The network contains these switches:

SwitchA - priority 61440, MAC 0000.0c63.ef20

SwitchB - priority 61440, MAC 0000.0c65.4ba6

SwitchC - priority 0, MAC 0000.0c65.a992

SwitchD - priority 4096, MAC 0000.0c60.0717

SwitchE - priority 0, MAC 0000.0c65.93ff

You have recently been hired to administer the network. The outgoing administrator told you that there is a problem with the root bridge but did not tell you which switch is the root.

Which of the following switches is the root bridge for the network? (Select the best answer.)

- A. SwitchB
- B. SwitchD
- C. SwitchA
- D. SwitchE
- E. SwitchC

Correct Answer: D

Explanation

SwitchE is the root bridge for the network. The root bridge is the switch with the lowest bridge ID (BID). The BID is composed of a 2-byte bridge priority and a 6-byte Media Access Control (MAC) address. The bridge priority is considered first in the determination of the lowest BID. The bridge priority can be set by issuing the spanning-tree priority value command, where value is a multiple of 4096 in the range from 0 through 61440; the default priority is 32768. SwitchC and SwitchE both have a priority of 0. When two or more switches have the lowest priority, the switch with the lowest MAC address becomes the root bridge. MAC addresses are written in hexadecimal format. In MAC addresses, numbers are lower than letters and the hexadecimal value A is lower than the hexadecimal value F. Because SwitchE has a lower MAC address than SwitchC, SwitchE is the root bridge.

Electing a root bridge is one of the first steps in the Spanning Tree Protocol (STP) loop-prevention process. STP also optimizes the network by finding the lowest cost paths, determines which port becomes the root port of each switch, and determines which ports become designated ports. STP determines port states through the use of the Spanning Tree Algorithm (STA). The STA uses the following procedure to determine port state:

- A root switch is elected, and each active port on the root switch is placed into the forwarding state.
- Each nonroot switch places the port with the lowest-cost path to the root switch, which is known as the root port, into the forwarding state.
- Ports with the lowest cost to nonroot switches, which are known as designated ports, are also placed into the forwarding state.

e All other ports are placed into the blocking state.

SwitchA is not the root bridge for the network, because it has the highest priority value, not the lowest priority value. Although link speed is somewhat relevant in determining the root port for a switch, link speed is irrelevant in determining the root bridge.

SwitchB is not the root bridge for the network; like SwitchA, SwitchB also has the highest priority value, not the lowest priority value. SwitchB contains redundant links to SwitchD, but redundant links are irrelevant in determining the root bridge. To avoid a switching loop, at least one of the redundant links between SwitchB and SwitchD will be blocked.

SwitchC is not the root bridge for the network. If the bridge priority of SwitchE were higher than 0, SwitchC would be the root bridge, because a priority of 0 is the lowest configurable priority value.

SwitchD is not the root bridge for the network. Although SwitchD has the lowest MAC address on the network, the bridge priority is considered first in the determination of the root bridge. If all of the switches on the network had the same bridge priority values, SwitchD would be the root bridge, because it has the lowest MAC address.

Original Exam B Question 55

Which of the following optional STP features reduces convergence time by immediately placing edge ports into a forwarding state? (Select the best answer.)

- A. BPDU filter

- B. BPDU guard
- C. root guard
- D. PortFast
- E. loop guard

Correct Answer: D

Explanation

PortFast is an optional Spanning Tree Protocol (STP) feature that reduces convergence time by immediately placing edge ports into a forwarding state. PortFast is recommended for ports that connect to end-user devices, such as desktop computers. In addition, PortFast cannot be used with loop guard. If PortFast is enabled on a loop guard-enabled port, loop guard will be automatically disabled. To enable PortFast, you should issue the spanning-tree portfast command from interface configuration mode.

Loop guard places inconsistent ports into the blocking state; it does not reduce convergence time by immediately placing edge ports into a forwarding state. Loop guard prevents a switch port from transitioning to the forwarding state when it stops receiving bridge protocol data units (BPDUs); this prevents switching loops from occurring. A port configured with loop guard that stops receiving BPDUs will be put into the loop-inconsistent state, as shown in the following output:

```
%SPANTREE-4-LOOPGUARDBLOCK: No BPDUs were received on port 0/1 in vlan 4. Moved to loop inconsistent state
```

After the port starts receiving BPDUs again, loop guard enables the port to transition through the normal STP states. To enable loop guard, you should issue the spanning-tree guard loop command from interface configuration mode.

Root guard is used to prevent newly introduced switches from being elected the new root; it does not reduce convergence time by immediately placing edge ports into a forwarding state. This allows administrators to maintain control over which switch is the root. When STP is used, the device with the lowest bridge priority is elected the root. If an additional device is added to the network with a lower priority than the current root, it will become the new root. However, this could cause the network to reconfigure in unintended ways. To prevent this, root guard can be applied. Root guard is applied on a per-port basis with the spanning-tree guard root command. If root guard is enabled on a loop guard-enabled port, loop guard will be automatically disabled.

BPDU guard is used to disable ports that erroneously receive BPDUs; it does not reduce convergence time by immediately placing edge ports into a forwarding state. BPDU guard is applied to edge ports that have PortFast enabled. Because PortFast automatically places ports into a forwarding state, a switch that has been connected to a PortFast-enabled port could cause switching loops. However, when BPDU guard is applied, the receipt of a BPDU on a port with BPDU guard enabled will result in the port being placed into a disabled state, which prevents loops from occurring. To enable BPDU guard, you should issue the spanning-tree bpduguard enable command from interface configuration mode.

The BPDU filter feature prevents PortFast-enabled ports from sending or receiving BPDUs; it does not reduce convergence time by immediately placing edge ports into a forwarding state. The BPDU filter behaves differently depending on whether it is enabled globally or only on specific ports. When enabled globally, the BPDU filter does not restrict BPDU flow on a PortFast interface until the interface is in an operational state. In addition, if a port receives a BPDU, it loses its PortFast operational status and the BPDU filter feature is automatically disabled for that port until it is operational. You can globally enable BPDU filtering by issuing the spanning-tree portfast bpdupfilter default command in global configuration mode. By contrast, when enabled for a specific port, the BPDU filter feature filters all inbound and outbound BPDUs for the port without regard to the operational state of the port. Configuring BPDU filtering at the port level effectively disables STP on that port and could result in a bridging loop. You can enable BPDU filtering on an individual port by issuing the spanning-tree bpdupfilter enable command in interface configuration mode.

Day 22 - Rapid STP

Included Exam B items: Q41

Original Exam B Question 41

Rapid PVST+ is a Cisco enhancement of which of the following IEEE protocols? (Select the best answer.)

- A. 802.1t
- B. 802.1w
- C. 802.1s
- D. 802.1D
- E. 802.10

Correct Answer: B

Explanation

Rapid Per-VLAN Spanning Tree Plus (PVST+) is a Cisco enhancement of the Institute of Electrical and Electronics Engineers (IEEE) 802.1w Rapid Spanning Tree Protocol (RSTP). Like PVST+, Rapid PVST+ provides an instance of RSTP for each configured virtual local area network (VLAN) as well as incorporates the rapid convergence abilities of RSTP. Faster convergence is achieved by replacing three port states with a single port state; the disabled, blocking, and listening states are merged into a new discarding port state.

PVST+, not Rapid PVST+, is a Cisco enhancement of IEEE 802.1D Spanning Tree Protocol (STP). STP maintains one spanning tree instance for the entire network, whereas PVST+ provides a separate spanning tree instance for each configured VLAN.

Rapid PVST+ is not a Cisco enhancement of IEEE 802.1s Multiple Spanning Tree (MST). Like Rapid PVST+, MST also uses IEEE 802.1w. MST can have up to 16 instances of RSTP and can combine several VLANs with the same topology into a common RSTP instance.

Rapid PVST+ is not a Cisco enhancement of IEEE 802.1t. IEEE 802.1t is an amendment to IEEE 802.1D and was later incorporated into the 802.1D standard.

Rapid PVST+ is not a Cisco enhancement of IEEE 802.1Q VLAN tagging. VLAN tagging provides a method for a large physical network to be broken down into smaller logical groupings, often used to separate departments within a company that share the same hardware resources. This logical segregation of departments provides a level of security between internal network segments.

Day 23 - EtherChannel

Included Exam B items: Q2

Original Exam B Question 2

Your company's network includes two Cisco switches, SwitchA and SwitchB. SwitchB has been configured with the correct trunk and EtherChannel settings.

SwitchA is being repurposed from a previous installation and must be reconfigured.

Configure SwitchA with the following parameters:

- * Configure switch ports FastEthernet 0/1 and FastEthernet 0/2 to use a Cisco-proprietary EtherChannel negotiation protocol.
 - Configure switch ports FastEthernet 0/1 and FastEthernet 0/2 as members of EtherChannel port-group 1, and ensure that they actively negotiate EtherChannel links.
- Configure the EtherChannel virtual interface to establish a trunk link with SwitchB, and ensure that the link uses a standards-based encapsulation protocol.
- Ensure that DTP is disabled on the EtherChannel virtual interface.

Source result: NetSim lab task - no separate answer/explanation is included in the uploaded Exam B PDF.

Days 24-25 - Dynamic Routing, RIP & EIGRP

Included Exam B items: Q45, Q69

Original Exam B Question 45

What AD is assigned to internal EIGRP routes by default? (Select the best answer.)

- A. 90
- B. 0
- C. 1
- D. 5
- E. 170

Correct Answer: A

Explanation

Internal Enhanced Interior Gateway Routing Protocol (EIGRP) routes are assigned an administrative distance (AD) of 90 by default. AD values are used to determine the routing protocol that should be preferred when multiple routes to a destination network exist. A routing protocol with a lower AD will be preferred over a route with a higher AD. The following list contains the most commonly used AD values:

Route Source AD

Directly connected route 0

Static route 1

EIGRP summary route

eBGP

Internal EIGRP

IGRP

OSPF

IS-IS

RIP

External EIGRP

iBGP

Unknown

EIGRP routes appear in the output of the show ip route command with a letter D, as shown in the following output:

Router1#show ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

.1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is not set

192.168.0.0/24 is variably subnetted, 9 subnets, 2 masks

jo 192.168.0.128/28 [90/2195456] via 192.168.0.37, 00:03:31, Serial0/1

The first number within the brackets is the AD. The second number within the brackets is the metric.

To modify the AD of incoming routes, you should issue the distance command. The syntax of the distance command is distance administrative-distance ip-address wildcard-mask [acl]. The ip-address and wildcard-mask parameters define the source of the route. The optional acl parameter, for which you can use the access control list (ACL) name or number, specifies the routes to which the AD should be applied. If no ACL is specified in the distance command, then the AD is applied to all routes received from the specified IP address or network.

Directly connected routes have an AD of 0. Therefore, directly connected routes are trusted over routes from any other source.

Static routes have an AD of 1. Therefore, static routes are more trusted than routes from any routing protocol. Static routes are optimal for routing networks that do not change often. To create a static route, you should issue the ip route command.

EIGRP summary routes have an AD of 5. Routes that are learned by EIGRP are called internal EIGRP routes and have an AD of 90. To modify the AD values used by EIGRP, you should issue the `distance eigrp internal external` command, where internal is the AD used for internal EIGRP routes and external is the AD used for external EIGRP routes.

Original Exam B Question 69

Which of the following statements is true regarding a floating static route? (Select the best answer.)

- A. A floating static route is used to provide link load balancing.
- B. A floating static route has a lower AD than a normal static route.
- C. A router always prefers a floating static route to a dynamically learned route.
- D. A floating static route is used to provide link redundancy.

Correct Answer: D

Explanation

A floating static route is used to provide link redundancy. When multiple routes to a network exist and a more specific route is not available, a router will choose the route with the lowest administrative distance (AD). Because a normal static route has a default AD of 1, a router will always prefer a normal static route over any other type of route. You can manually assign a static route a higher AD than 1 to prevent a router from always choosing the normal static route as the best path to a destination network. By assigning a floating static route a higher AD than another route, you are able to create a static route that will be used only when routes with a lower AD are no longer available. For example, if a router's primary path to a remote office is a dynamically learned route and a floating static route with a higher AD is configured to use a specified exit interface as a backup path, the router will use only the primary route to reach the remote office. The dynamically learned route is preferred over the floating static route because the floating static route has a higher AD than the dynamically learned route. However, if the dynamically learned route becomes unavailable, the router will search its routing table for an available path with the lowest AD. In this example, the router will use the floating static route to forward packets destined to the remote office to the exit interface specified in the floating static route when the dynamically learned route becomes unavailable.

The syntax of the `ip route` command is `ip route prefix mask {ip-address | interface} [distance]`, where prefix is the network address of the destination network, mask is the subnet mask of the destination network, ip-address is the Internet Protocol (IP) address of the next-hop router, interface is the local interface to which the packets should be sent, and distance is the AD of the route. For example, to create a floating static route that could be used whenever the primary static route goes down, you could issue the `ip route 10.2.2.0 255.255.255.0 192.168.1.1 10` command.

A router will not always prefer a floating static route to a dynamically learned route. Because an administrator can arbitrarily assign an AD to a floating static route, a router will prefer a floating static route only if it has a lower AD than a dynamically learned route to the same destination network.

Likewise, a router will not always prefer a dynamically learned route to a floating static route unless the dynamically learned route has an AD lower than a floating static route to the same destination network. For example, to create a floating static route that would be used as a backup link to an internal Enhanced Interior Gateway Routing Protocol (EIGRP) route, you might issue the following command:

```
ip route 192.168.171.0 255.255.255.0 192.168.1.5 91
```

Assuming that the default EIGRP AD of 90 is used for the primary link, this floating static route would be used if the EIGRP route to 192.168.171.0/24 were to become unavailable. The floating static route would then direct traffic to the next hop address 192.168.1.5. However, if the AD of the floating static route were set to a value of 89, it would always be preferred over the EIGRP route.

A floating static route is not used for link load balancing. Load balancing is possible if multiple paths to a destination network exist with equal ADs and if cost values exist. Because a floating static route has a higher AD than the primary path to a destination network, a router will not use a floating static route unless the primary path becomes unavailable.

Days 26-28 - OSPF

Included Exam B items: Q14, Q22, Q24, Q54, Q70

Original Exam B Question 14

You administer the network in the exhibit. RouterA's Serial 0/0 interface is directly connected to the ISP and should provide the default route for the entire network. No static routes are configured on RouterB.

You issue the show running-config command on RouterA and receive the following partial output:

```
router ospf 100
 log-adjacency-changes
 network 10.10.10.0 0.0.0.3 area 0
 network 123.45.67.89 0.0.0.3 area 0
 default-information originate
!
ip route 0.0.0.0 0.0.0.0 10.10.10.2
```

Which of the following statements is true? (Select the best answer.)

- A. The network 10.10.10.0 0.0.0.3 area 0 command contains invalid syntax.
- B. Traffic from RouterB to the Internet will be forwarded through RouterA.
- C. The static route is misconfigured on RouterA.
- D. RouterA is not configured to advertise a gateway of last resort.

Correct Answer: C

Explanation

The static route is misconfigured on RouterA; therefore, RouterB will not be configured with a gateway of last resort. The ip route 0.0.0.0 0.0.0.0 10.10.10.2 command in this scenario configures RouterA to use the FastEthernet 1/0 interface on RouterB as a gateway of last resort instead of the Serial 0/0 interface on RouterA, which is the interface that is connected to the Internet service provider (ISP). In addition, the default-information originate command is configured on RouterA. Issuing the default-information originate command in Open Shortest Path First (OSPF) router configuration mode will cause RouterA to inject its statically configured default route into OSPF, thereby advertising the default route to neighboring routers and propagating neighboring routers with a gateway of last resort. However, in this case, the OSPF process on RouterB will not place the route into its routing table, because the next-hop address advertised by RouterA is the address of one of RouterB's broadcast interfaces. If RouterA and RouterB were connected over a serial link, RouterB would place the advertised default route into its routing table and would create a routing loop. Traffic from RouterB to the Internet will not be forwarded through RouterA. RouterA is not advertising its default route to RouterB in this scenario. In addition, RouterB is not configured with its own gateway of last resort.

RouterA is configured to advertise a gateway of last resort. The default-information originate command configures an OSPF router to inject its default route into OSPF as an external route, thereby advertising its default route to neighboring routers.

The network 10.10.10.0 0.0.0.3 area 0 command does not contain invalid syntax. To configure a particular interface or network to connect to a specific OSPF area, you should issue the network ip-address wildcard-mask area area-id command in OSPF router configuration mode. By using the ip-address and wildcard-mask parameters, you can specify a router interface, a single network, or a group of networks. In this scenario, the network 10.10.10.0 0.0.0.3 area 0 command correctly configures OSPF to advertise the 10.10.10.0/30 network.

Original Exam B Question 22

You issue the ip ospf network non-broadcast command on an interface.

Which of the following statements is correct regarding how OSPF operates on the interface? (Select the best answer.)

- A. The neighbor command is required to establish adjacencies.
- B. The hello timer is set to 10 seconds, and the dead timer is set to 40 seconds.
- C. DR and BDR elections are not performed.
- D. Multicast updates are sent.

Correct Answer: A

Explanation

The neighbor command is required to establish adjacencies on Open Shortest Path First (OSPF) nonbroadcast networks. There are five OSPF network types:

- Broadcast
- Nonbroadcast

Point-to-point

- Point-to-multipoint broadcast
- Point-to-multipoint nonbroadcast

Nonbroadcast and point-to-multipoint nonbroadcast networks do not allow multicast packets. To configure OSPF to send unicast updates, you must configure neighbor routers with the neighbor command. Broadcast, point-to-point, and point-to-multipoint broadcast networks allow multicast packets, so manual configuration of neighbor routers with the neighbor command is not required. On broadcast networks, designated router (DR) and backup designated router (BDR) elections are performed. By default, the hello timer is set to 10 seconds and the dead timer is set to 40 seconds. To configure an OSPF broadcast network, you should issue the ip ospf network broadcast command. The OSPF broadcast network type is enabled by default on Fiber Distributed Data Interface (FDDI) and Ethernet interfaces, including Fast Ethernet and Gigabit Ethernet interfaces.

On nonbroadcast networks, DR and BDR elections are performed. By default, the hello timer is set to 30 seconds and the dead timer is set to 120 seconds. To configure an OSPF nonbroadcast network, which is also called a nonbroadcast multiaccess (NBMA) network, you should issue the ip ospf network non-broadcast command.

On point-to-point networks, DR and BDR elections are not performed. By default, the hello timer is set to 10 seconds and the dead timer is set to 40 seconds. To configure an OSPF point-to-point network, you should issue the ip ospf network point-to-point command. The OSPF point-to-point network type is enabled by default on High-Level Data Link Control (HDLC) and Point-to-Point Protocol (PPP) serial interfaces.

OSPF point-to-multipoint broadcast networks operate just like OSPF point-to-point networks except the hello timer is set to 30 seconds and the dead timer is set to 120 seconds by default. To configure an OSPF point-to-multipoint broadcast network, you should issue the ip ospf network point-to-multipoint command.

OSPF point-to-multipoint nonbroadcast networks operate just like OSPF point-to-multipoint broadcast networks except that multicasts cannot be sent; therefore, manual configuration of neighbor routers with the neighbor command is required so that OSPF sends unicast updates. To configure an OSPF point-to-multipoint nonbroadcast network, you should issue the ip ospf network point-to-multipoint non-broadcast command.

Original Exam B Question 24

If the OSPF router ID of a router is not manually configured, in what order is the router ID determined? (Select the best answer.)

- A. first by the OSPF priority value, then by the highest IP address among loopback interfaces, then by the highest IP address among physical interfaces
- B. first by the lowest IP address among loopback interfaces, then by the lowest IP address among physical interfaces
- C. first by the highest IP address among physical interfaces, then by the highest IP address among loopback interfaces
- D. first by the lowest IP address among physical interfaces, then by the lowest IP address among loopback interfaces
- E. first by the OSPF priority value, then by the lowest IP address among physical interfaces, then by the lowest IP address among loopback interfaces
- F. first by the highest IP address among loopback interfaces, then by the highest IP address among physical interfaces

Correct Answer: F

Explanation

If the Open Shortest Path First (OSPF) router ID of a router is not manually configured, the router ID is determined first by the highest Internet Protocol (IP) address among loopback interfaces, then by the highest IP address among physical interfaces if no loopback interfaces have been configured. To manually configure a router ID for an OSPF router, you should issue the router-id value command in router configuration mode, where value is a 32-bit value in dotted decimal notation. If the router ID is not manually configured, the router ID will be the highest loopback IP address configured on a router.

If a loopback IP address is not configured, the router ID will be the highest IP address among configured interfaces on the router. Because loopback interfaces are logical, not physical, a loopback interface can never fail. For this reason, configuring a loopback interface is recommended for OSPF operation, because loopback interfaces are more reliable than physical interfaces. The highest IP address among physical interfaces will only be used for the router ID if the router ID is not manually configured and if a loopback address is not configured. The router ID is not determined by the lowest IP address among interfaces.

The OSPF priority value is not used for determining the router ID. The priority value is used for electing the designated router (DR) and backup designated router (BDR) for an OSPF multiaccess network. The DR is typically the router with the highest OSPF priority. The DR and BDR generate link-state advertisements (LSAs) that contain OSPF routing information. The router with the highest priority becomes the DR, and the router with the second-highest priority becomes the BDR. By default, the OSPF priority is 1. To change the OSPF priority, issue the `ip ospf priority value` command, where value is a number from 0 through 255; a router with a priority of 0 will never be elected to be the DR or BDR. If priorities are equal between two or more routers, the router with the highest router ID will be elected to become the DR.

Original Exam B Question 54

You are attempting to configure OSPF between RouterA and RouterB on your network. However, the routers are unable to form an OSPF adjacency. You issue the `show ip ospf interface fastethernet 0/0` command on each router and receive the following output:

```
RouterA#show ip ospf interface fastethernet 0/0
FastEthernet 0/0 is up, line protocol is up
Internet Address 10.1.1.13/30, Area 0
Process ID 101, Router ID 192.168.15.237, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 192.168.15.237, Interface address 10.1.1.13
No Backup Designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:07
Index 1/1, flood queue length 0
Next 0x0(0)/0xe(0)
Last flood scan length is 2, maximum is 2
Last flood scan time is 0 msec, maximum is 4 msec
Neighbor Count is @, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)

RouterB#show ip ospf interface fastethernet 0/0
FastEthernet 0/0 is up, line protocol is up
Internet Address 10.1.1.14/30, Area 1
Process ID 102, Router ID 192.168.15.237, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 192.168.15.237, Interface address 10.1.1.14
No Backup Designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:05
Index 1/1, flood queue length 0
Next 0x0(0)/0xe(e)
Last flood scan length is 2, maximum is 2
Last flood scan time is 0 msec, maximum is 4 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)

Why are the routers unable to form an adjacency? (Select 2 choices.)
A. The router IDs must not match.
B. The dead timer value must match the hello timer value.
C. The OSPF process IDs must match.
D. The IP addresses are configured with the wrong subnet mask.
E. The OSPF areas must match.
F. The cost must be higher than 1.
```

Correct Answer: A, E

Explanation

The routers are unable to form an adjacency for two reasons: the Open Shortest Path First (OSPF) areas must match, and the router IDs must not match. Only OSPF routers in the same hierarchical area form adjacencies. To configure an OSPF area, you should issue the `network address mask area area` command in OSPF router configuration mode.

Although you can configure a multiarea OSPF topology, the OSPF areas on neighboring interfaces must match in order for two OSPF routers to form an adjacency. In this scenario, RouterB's FastEthernet 0/0 interface is operating in Area 1 and is directly connected to RouterA's FastEthernet 0/0 interface, which is operating in Area 0. Area 0 is also known as the backbone area. If you are not using virtual links, any nonbackbone OSPF areas that you configure must border Area 0. On smaller networks, single-area OSPF configurations also offer the benefit of generating fewer link-state advertisements (LSAs).

Router IDs must be unique; a router will ignore hello packets that are marked with its own router ID. To manually configure the router ID, you should issue the `router-id ip-address` command in router configuration mode, where `ip-address` is a 32-bit value in dotted decimal notation. If the router ID is not manually configured, the router ID is the highest loopback Internet Protocol (IP) address configured on a router. If a loopback IP address is not configured, then the router ID is the highest IP address among configured interfaces on the router.

Whereas Enhanced Interior Gateway Routing Protocol (EIGRP) process IDs must match when EIGRP is used, OSPF process IDs do not have to match when OSPF is used. The OSPF process ID is locally significant to the router and can be any positive integer in the range from 1 through 65535. You can specify the OSPF process ID by issuing the `router ospf process-id` command when you configure a router for OSPF.

The cost does not have to be higher than 1. By default, OSPF uses the bandwidth of a link to determine the cost. The higher the bandwidth, the lower the cost. To calculate the cost, divide 100,000,000 by the bandwidth in bits per second (bps). The FastEthernet 100-megabits per second (Mbps) link between RouterA and RouterB would have a default cost of 1, because $100,000,000 \div 100,000,000 = 1$. To manually configure the OSPF cost of a link, you should issue the `ip ospf cost cost` command in interface configuration mode.

The IP addresses for RouterA and RouterB are within the same subnet. Therefore, the subnet mask is correct, and the two routers should be able to communicate. A /30 subnet mask allows for two hosts per subnet. In this example, 10.1.1.12/30 is the network address, 10.1.1.13/30 and 10.1.1.14/30 are valid host addresses, and 10.1.1.15/30 is the broadcast address.

The dead timer value should not match the hello timer value. In order for OSPF routers to establish an adjacency, the dead timer on one router should match the dead timer on the other router and the hello timer on one router should match the hello timer on the other router. Both RouterA and RouterB are set to a hello interval of 10 seconds and a dead interval of 40 seconds, so the OSPF timers are configured correctly. The hello timer is used to specify the amount of time between sending hello packets, and the dead timer is used to specify the amount of time to wait for hello packets before declaring a neighbor to be down. By default, the hello timer is set to 10 seconds on point-to-point and broadcast links and 30 seconds on nonbroadcast multiaccess (NBMA) links. The dead timer is set to four times the hello timer value by default. To manually configure the hello timer interval, you should issue the `ip ospf hello-interval seconds` command in interface configuration mode. To manually configure the dead timer interval, you should issue the `ip ospf dead-interval seconds` command in interface configuration mode.

Original Exam B Question 70

Which of the following OSPF network types have a default hello timer of 10 seconds and a default dead timer of 40 seconds? (Select 2 choices.)

- A. broadcast
- B. point-to-multipoint nonbroadcast
- C. nonbroadcast
- D. point-to-point
- E. point-to-multipoint

Correct Answer: A, D

Explanation

The broadcast and point-to-point Open Shortest Path First (OSPF) network types have a default hello timer of 10 seconds and a default dead timer of 40 seconds. The nonbroadcast, point-to-multipoint, and point-to-multipoint nonbroadcast OSPF network types have a default hello timer of 30 seconds and a dead timer of 120 seconds.

The hello timer is used to specify the amount of time between hello packets. The dead timer is used to specify the amount of time to wait before declaring a neighbor to be down. In order for OSPF routers to establish an adjacency, the hello timer on one router should match the hello timer on the other router, and the dead timer on one router should match the dead timer on the other router. The dead timer is set to four times the hello timer value by default.

There are five OSPF network types:

- Broadcast
- Nonbroadcast
- Point-to-point
- Point-to-multipoint broadcast
- Point-to-multipoint nonbroadcast

The OSPF broadcast networktype is enabled by default on Fiber Distributed Data Interface (FDDI) and Ethernet interfaces, including Fast Ethernet and Gigabit Ethernet interfaces. On broadcast networks, multicast updates are sent, so manual configuration of neighbor routers with the neighbor command is not required. To configure an OSPF broadcast network, you should issue the ip ospf network broadcast command.

The OSPF point-to-point network type is enabled by default on High-Level Data Link Control (HDLC) and Point-to-Point Protocol (PPP) serial interfaces. On point-to-point networks, designated router (DR) and backup designated router (BDR) elections are not performed. Multicast updates are sent, so manual configuration of neighbor routers with the neighbor command is not required. To configure an OSPF point-to-point network, you should issue the ip ospf network point-to-point command.

On nonbroadcast networks, DR and BDR elections are performed. Nonbroadcast networks do not allow multicasts; therefore, manual configuration of neighbor routers with the neighbor command is required so that OSPF sends unicast updates. To configure an OSPF nonbroadcast network, which is also called a nonbroadcast multiaccess (NBMA) network, you should issue the ip ospf network non-broadcast command.

OSPF point-to-multipoint networks operate just like OSPF point-to-point networks except the hello timer is set to 30 seconds and the dead timer is set to 120 seconds by default. To configure an OSPF point-to-multipoint network, you should issue the ip ospf network point-to-multipoint command.

OSPF point-to-multipoint nonbroadcast networks operate just like OSPF point-to-multipoint broadcast networks except that multicasts cannot be sent; therefore, manual configuration of neighbor routers with the neighbor command is required so that OSPF sends unicast updates. To configure an OSPF point-to-multipoint nonbroadcast network, you should issue the ip ospf network point-to-multipoint non-broadcast command.

Day 29 - First Hop Redundancy Protocols

Included Exam B items: Q25, Q46

Original Exam B Question 25

Drag each description to its corresponding FHRP.

Correct Answer: GLBP - has one active virtual gateway and multiple active virtual forwarders

HSRP - has one active router and one standby router

VRRP - has one master router and multiple backup routers

Explanation

Gateway Load Balancing Protocol (GLBP) is a First-Hop Redundancy Protocol (FHRP) that also provides load balancing. GLBP enables you to configure multiple routers as a GLBP group; the routers in the group receive traffic sent to a virtual Internet Protocol (IP) address that is configured for the group. Each GLBP group contains an active virtual gateway (AVG) that is elected based on which router is configured with the highest priority value, or with the highest IP address if multiple routers are configured with the highest priority value. The other routers in the GLBP group are configured as primary or secondary active virtual forwarders (AVFs). Up to four primary AVFs can be configured in a GLBP group, and the primary AVFs can participate in forwarding traffic. Consequently, multiple routers can be used simultaneously to provide load balancing for the GLBP group.

Hot Standby Router Protocol (HSRP) is an FHRP that is defined in Request for Comments (RFC) 2281. Similar to GLBP, HSRP can be used to provide backup router coverage if the primary gateway becomes unavailable. Multiple routers are assigned to an HSRP group, and the routers function as a single gateway. An HSRP group contains one active router and one standby router. The active router is the router with the highest priority value, and the standby router is the router with the second-highest priority value. Other routers in the HSRP group are in the listen state. If the active router fails, the standby router will assume the active router role and a new standby router will be elected.

Virtual Router Redundancy Protocol (VRRP) is also an FHRP. Routers are assigned to a VRRP group, and the group functions as a single gateway for clients. A VRRP group has one master router, which is the router with the highest priority value. All other routers in the VRRP group are backup routers. A virtual Media Access Control (MAC) address is used to identify the VRRP group to clients. The virtual MAC address for VRRP groups is in the form of 0000.5e00.01xx, where xx is a hexadecimal value identifying the VRRP group number. For example, VRRP Group 1 would be identified by the virtual MAC address 0000.5e00.0101.

Original Exam B Question 46

Which of the following is an IETF-standard FHRP that can use object tracking and preemption to provide Layer 3 failover? (Select the best answer.)

- A. VRRP
- B. HSRP
- C. GLBP
- D. LACP

Correct Answer: A

Explanation

Virtual Router Redundancy Protocol (VRRP) is an Internet Engineering Task Force (IETF)-standard First-Hop Redundancy Protocol (FHRP) that can use object tracking and preemption to provide Layer 3 failover. FHRPs are protocols that are used to provide Layer 3 gateway redundancy, such as failover and load balancing. Providing Layer 3 redundancy ensures that hosts on a local area network (LAN) will have a backup path to external networks should a primary path fail or become too congested to forward traffic. Layer 3 devices in an FHRP configuration typically share a virtual Internet Protocol (IP) address that is then configured as the default gateway on each host for which the device is to forward traffic. The FHRP devices might also share a virtual Media Access Control (MAC) address or multiple virtual MAC addresses, depending on the protocol. FHRPs typically use a priority system to elect a primary Layer 3 forwarding device, which is known as an active virtual gateway (AVG), an active router, or a master router, depending on the protocol. The same priority system elects either a single or multiple backup-forwarding devices.

VRRP can be configured to use object tracking to influence the priority of a router in a group and therefore force the election of a different master router when certain conditions are met. When combined with VRRP preemption, which enables a VRRP router to automatically assume the master router role when priority values change, object tracking enables VRRP to adjust the priority of a router based on the line protocol status of a specific interface or the availability of a given route to a destination. For example, if RouterA and RouterB in a VRRP configuration had different paths to the Internet, VRRP could be configured to monitor RouterA's outbound interface and to automatically set RouterA's VRRP priority to a value lower than RouterB's if RouterA's outbound interface were to go down. RouterB would then become the master router, and Layer 3 traffic would be forwarded through its outbound interface instead of through RouterA's.

Gateway Load Balancing Protocol (GLBP) is a Cisco-proprietary FHRP, not an IETF standard. However, GLBP does support object tracking and preemption. GLBP is different from both VRRP and Hot Standby Router Protocol (HSRP) in that it is, by default, capable of load balancing traffic between all routers in a GLBP group. VRRP and HSRP are primarily failover protocols. GLBP elects an AVG and up to four primary active virtual forwarders (AVFs). The routers in a GLBP group receive traffic sent to a virtual IP address that is configured for the group. Each GLBP group contains an AVG that is elected based on which router is configured with the highest priority value or the highest IP address value if multiple routers are configured with the highest priority value. The other routers in the GLBP group are configured as primary or secondary AVFs. The AVG in a GLBP group assigns a virtual MAC address to up to four primary AVFs; all other routers in the group are considered secondary AVFs and are placed in the listen state. When the AVG receives ARP requests that are sent to the virtual IP address for the GLBP group, the AVG responds with different virtual MAC addresses. This provides load balancing, because each of the primary AVFs will participate by forwarding a portion of the traffic sent to the virtual IP address.

HSRP is a Cisco-proprietary FHRP, not an IETF standard. However, like VRRP, HSRP is capable of using object tracking and preemption to modify the priority of the HSRP active router and force the standby router to take over if a specific interface goes down or a path to the destination becomes unavailable.

Link Aggregation Control Protocol (LACP) is not an FHRP. LACP is an Institute of Electrical and Electronics Engineers (IEEE) protocol that is used to enable link aggregation on EtherChannel links.

Day 30 - TCP & UDP

Included Exam B items: Q26

Original Exam B Question 26

Which of the following relies on a three-way handshake in order to provide connection-oriented, reliable data transfer between networked computers? (Select the best answer.)

- A. ARP
- B. RARP
- C. UDP
- D. DNS
- E. TCP

Correct Answer: E

Explanation

Transmission Control Protocol (TCP) relies on a three-way handshake in order to provide connection-oriented, reliable data transfer between networked computers. When the TCP process on a host initiates a connection to the TCP process on another host, it sends an empty Protocol Data Unit (PDU) with a TCP header that contains a SYN bit, which signifies that a synchronization request is being initiated. In addition, the header will contain the value of the first sequence number for the newly created communication channel. The TCP process at the destination host will respond with a similarly empty PDU that specifies its own SYN bit and starting sequence number. In addition, the TCP header will include an ACK bit, which acknowledges the receipt of the previous PDU. The header will also contain the value of the next sequence number that the process is expecting from the sender. The value of the next sequence number is typically one digit higher than the last sequence number that was transmitted, unless there was a transmission error. In this way, the TCP process on the initiating host knows which of its transmissions have been received. When the TCP process on the initiating host receives the acknowledgment PDU, it replies with its own acknowledgment PDU. This acknowledgment to the destination host contains the ACK bit and the value of the next sequence number that the originating TCP process is expecting from the destination TCP process. This process ensures that both TCP processes are synchronized and is referred to as the TCP three-way handshake. TCP is considered reliable because it offers features such as error detection and correction, flow control, sequencing, and acknowledgments.

User Datagram Protocol (UDP) does not rely on a three-way handshake. In addition, UDP does not provide connection-oriented, reliable data transfer between networked computers. UDP provides connectionless, unreliable data transfer between networked computers. The term connectionless means that UDP does not establish a communications session before sending data. UDP is considered unreliable because a destination computer does not send an acknowledgment that the data was received.

Reverse Address Resolution Protocol (RARP) does not rely on a three-way handshake. In addition, RARP does not provide connection-oriented, reliable data transfer between networked computers. RARP translates Media Access Control (MAC) addresses to Internet Protocol (IP) addresses.

Address Resolution Protocol (ARP) does not rely on a three-way handshake. In addition, ARP does not provide connection-oriented, reliable data transfer between networked computers. ARP, which is the reverse of RARP, is used to translate IP addresses to MAC addresses.

Domain Name System (DNS) does not rely on a three-way handshake in order to provide connection-oriented, reliable data transfer between networked computers. DNS translates host names to IP addresses.

Days 31-33 - IPv6

Included Exam B items: Q16, Q40, Q61

Original Exam B Question 16

Which of the following statements is true regarding link-local addresses? (Select the best answer.)

- A. They are used for ULA.
- B. They must never be used as destination addresses.
- C. They are not routable.
- D. They must never be used as source addresses.
- E. They must be globally unique.

Correct Answer: C

Explanation

Link-local addresses are an Internet Protocol version 6 (IPv6) address type that is not routable. IPv6 link-local addresses are unicast addresses used for one-to-one communication. Because link-local unicast addresses are unique only on the local segment, they need not be globally unique and are not routable. An IPv6-capable host typically creates a link-local unicast address automatically at startup. Link-local unicast addresses are used for neighbor discovery and for environments in which no router is present to provide a routable IPv6 prefix. These addresses always begin with FE8, FE9, FEA, or FEB. By contrast, a global unicast address, which is also referred to as an aggregatable global address, must be globally unique. Global unicast addresses are designed to minimize the size of Internet routing tables. A global unicast address contains three distinct parts:

- The Global Routing Prefix · identifies the public portion of the address, as assigned by a service provider
- The Site-Level Aggregator (SLA) · identifies the site or group of sites associated with the address
- The Interface ID · identifies the address assigned to the interface of the network device associated with the address

The Global Routing Prefix is a 48-bit field that is defined by the Internet service provider (ISP). The SLA is a 16-bit field that identifies a site and is analogous to a subnet in IP version 4 (IPv4). The Interface ID is a 64-bit field that must be globally unique; therefore, it typically contains the Media Access Control (MAC) address of the originating device in extended unique identifier (EUI)-64 format. Because there is an inherent hierarchy in the aggregatable global address scheme, these addresses lend themselves to simple consolidation, which greatly reduces the complexity of Internet routing tables.

Multicast addresses, not link-local addresses, must never be used as source addresses. The prefix FFO0::/8 is used for multicast addresses, which are used for one-to-many communication. IPv6 addresses in the FFO0::/8 range begin with the characters FFO0 through FFFF. However, certain address ranges are used to indicate the scope of the multicast address. The following IPv6 multicast scopes are defined:

- FFO1::/16 · node-local
- FFO2::/16 · link-local
- FFO5::/16 · site-local
- FFO8::/16 · organization-local
- FFOE::/16 · global

Unicast site-local addresses, not link-local addresses, are used for unique local addressing (ULA). Addresses with the IPv6 prefix FC00::/7 are used for private addressing in a fashion similar to the use of Request for Comments (RFC) 1918 addresses on IPv4 networks. The IPv6 prefix FC00::/7 refers to addresses that begin with the characters FCO0 through FDFF. RFC 4193 outlines the best practices for private IPv6 addressing.

Original Exam B Question 40

You are migrating your company's IPv4 network to IPv6. The company has decided to use the 2001:DB8:A::/120 subnet throughout the network. You have been asked to configure RouterA and RouterB with IPv6 addresses on each end of the point-to-point link. RouterA should be assigned the next-to-last host address in the company's address range, and RouterB should be assigned the last host address. In order to

conserve address space, you have been asked to use the /126 subnet.

Which of the following commands should you issue to configure the interface on RouterB? (Select the best answer.)

- A. ipv6 address 2001:DB8:A::FFFE/126
- ☐ B. ipv6 address 2001:DB8:A::00FE/126
- C. ipv6 address 2001:DB8:A::FFFF/126
- D. ipv6 address 2001:DB8:A::00FF/126

Correct Answer: D

Explanation

You should issue the ipv6 address 2001:DB8:A::00FF/126 command to configure the interface on RouterB. A /126 prefix uses 126 bits for the network portion of the address and 2 bits for the host portion of the address. With 2 bits available for host addressing, four addresses can exist. The company's network is using the 2001:DB8:A::/120 subnet, which includes the 256 Internet Protocol version 6 (IPv6) addresses from 2001:DB8:A::0000 to 2001:DB8:A::00FF. In this scenario, RouterA should be assigned the next-to-last host address in the company's address range and RouterB should be assigned the last host address. Therefore, RouterA should be assigned the address 2001:DB8:A::00FE and RouterB should be assigned the address 2001:DB8:A::00FF. Unlike IP version 4 (IPv4), IPv6 does not reserve the last address in a subnet for the broadcast address, because IPv6 does not use broadcasts. Therefore, the last address in a subnet can be used for host addressing. You should issue the ipv6 address 2001:DB8:A::00FE/126 command to configure the interface on RouterA, not RouterB. In this scenario, RouterA should be assigned the next-to-last host address in the company's address range.

You should not issue the ipv6 address 2001:DB8:A::FFFE/126 command or the ipv6 address 2001:DB8:A::FFFF/126 command. The IPv6 addresses 2001:DB8:A::FFFE/126 and 2001:DB8:A::FFFF/126 are not the last addresses in the company's address range.

Original Exam B Question 61

You issue the following command on a Cisco router in global configuration mode:

```
ipv6 route 2001:db8:a::/32 fastethernet 0/1 2001:db8:b::1
```

Which of the following have you created? (Select the best answer.)

- A. a fully specified static route
- B. a floating static route
- C. a directly attached static route
- D. a recursive static route

Correct Answer: A

Explanation

You have created a fully specified Internet Protocol version 6 (IPv6) static route if you issue the ipv6 route 2001:db8:a::/32 fastethernet 0/1 2001:db8:b::1 command on a Cisco router in global configuration mode. A fully specified static route is an IPv6 static route in which the destination network, outbound interface, and next-hop IPv6 address are all configured directly. In this scenario, the IPv6 prefix of 2001:db8:a::/32 represents the destination network. The router's FastEthernet 0/1 interface is the outbound interface. The IPv6 address of 2001:db8:b::1 is the next-hop IPv6 address. Fully specified static routes are most often used when the outbound interface is multiaccess and could therefore be configured with multiple next-hop addresses. The next-hop address that is specified in the command must be directly connected to the outbound interface.

There are three other types of IPv6 static routes: directly attached static routes, recursive static routes, and floating static routes. A directly attached static route specifies the destination IPv6 network and the outbound interface. For example, the ipv6 route 2001:db8:a::/32 fastethernet 0/1 command configures a directly attached static route. When a directly connected static route is configured, the router assumes that any packet that matches the destination network is reachable through the specified outbound interface. Therefore, the packet's full destination address is used as the IPv6 next-hop address.

A recursive static route specifies the destination IPv6 network and the IPv6 next-hop address only. For example, the ipv6 route 2001:db8:a::/32 2001:db8:a::1 command configures the router to resolve all IPv6 addresses in the 2001:db8:a::/32 prefix through the next hop that has been assigned the IPv6 address of 2001:db8:a::1. The router assumes the outbound interface to be the interface to which the

next hop is either directly or indirectly connected. In other words, the next-hop IPv6 address must be resolvable through the outbound interface.

A floating static route can be comprised of any of the other three types of IPv6 static routes. A floating static route is typically used as a backup route when the primary route goes down. Floating static routes are typically configured with an administrative distance (AD) value that is numerically higher than the AD of the primary route. For example, the `ipv6 route 2001:db8:a::/32 5` command will install a static route with an AD of 5. If a static route with the default AD of 1 were to go down, the floating static route could be used as a backup route to the destination network.

Days 34-35 - Access Control Lists (ACLs)

Included Exam B items: Q47, Q51, Q58

Original Exam B Question 47

Your supervisor informs you that several unauthorized employees in LAN A are downloading large files from FTP sites on the external network. Your supervisor has asked you to block all FTP access for LAN A. FTP access should be available to other users. The subnet corresponding to LAN A is 10.1.1.0/24.

Which of the following access list commands could you issue on RouterA to create an ACL that can be used to achieve your goal? (Select the best answer.)

- A. access-list 1 deny tcp 10.1.1.0 0.0.0.255 any eq ftp
access-list 1 permit ip any any
- B. access-list 101 deny tcp 10.1.1.0 0.0.0.255 any eq 21
access-list 101 permit ip any any
- C. access-list 101 deny tcp any 10.1.1.0 0.0.0.255 eq ftp
access-list 101 permit ip any any
- D. access-list 1 deny tcp any 10.1.1.0 0.0.0.255 eq 21
access-list 1 permit ip any any

Correct Answer: B

Explanation

You could issue the access-list 101 deny tcp 10.1.1.0 0.0.0.255 any eq 21 command and the access-list 101 permit ip any any command on RouterA to create an access list that can be used to prohibit File Transfer Protocol (FTP) connections from LAN A to the external network without limiting other access. You could use either the ftp keyword or the 21 keyword at the end of the access-list command to specify FTP access. The access-list 101 permit ip any any command is required in access control list (ACL) 101 because an implicit deny statement at the end of every ACL denies any packets that have not been explicitly permitted. ACL 101 could then be applied to the FastEthernet 0/1 interface in the inbound direction to filter all FTP access for LAN A.

Extended ACLs are created by using the following command syntax:

```
access-list access-list-number [dynamic dynamic-name [timeout {deny | permit}
protocol source source-wildcard [operator [port]] destination destination-wildcard [operator [port]
[established] [precedence precedence] [tos tos] [log | log-input] [time-range time-range-name]
```

Issuing the access-list 101 deny tcp any 10.1.1.0 0.0.0.255 eq ftp command on RouterA would deny FTP access to LAN A, not from LAN A.

You should not issue the access-list 1 deny tcp 10.1.1.0 0.0.0.255 any eq ftp or access-list 1 deny tcp any 10.1.1.0 0.0.0.255 eq 21 commands, because access list number 1 indicates a standard access list. There are two classes of ACLs: standard and extended. Whereas standard ACLs can be used to filter only on source Internet Protocol (IP) addresses, extended ACLs can be used to filter on source and destination IP addresses, protocols, and ports. Because you are filtering on a specific port, in this case the default FTP port 21, you should use an extended ACL. Access list number ranges 1-99 and 1300-1999 are valid for standard IP ACLs. Access list number ranges 100-199 and 2000-2699 are valid for extended IP ACLs.

Standard ACLs are created by using the following command syntax:

```
access-list access-list-number {deny | permit} {host | source source-wildcard | any}
```

Original Exam B Question 51

You administer the network shown in the following exhibit:

RouterB has been assigned an IP address of 192.168.1.50 on the same /30 subnet as the RouterA interface that connects to RouterB.

After a failed attempt to telnet from RouterA to RouterB, you connect to the console of RouterB and issue the show access-lists command. The command produces the following output:

```
RouterB#show access-lists
Standard IP access list 10
10 permit host 192.168.1.34 (0 matches)
20 permit host 192.168.1.58 (5 matches)
Extended IP access list 101
```

```
10 permit tcp host 192.168.1.33 any eq telnet (0 matches)
20 permit tcp host 192.168.1.49 any eq telnet (0 matches)
You then issue the show running-config | section line command on RouterB and receive the following output:
RouterB#show running-config | section line
line con 0
line aux 0
line vty 04
access-class 10 in
Why is RouterA unable to telnet to RouterB? (Select the best answer.)
A. ACL 101 on RouterA is applied to the wrong interface.
B. ACL 101 on RouterB does not permit Telnet traffic.
C. ACL 10 on RouterB does not permit RouterA's IP address.
D. ACL 101 on RouterA does not permit RouterB's IP address.
```

Correct Answer: C

Explanation

RouterA cannot telnet to RouterB because access control list (ACL) 10 on RouterB does not permit RouterA's Internet Protocol (IP) address. You can determine the configuration of ACLs on RouterB by issuing the show access-lists command, as shown in the following output:

```
RouterB#show access-lists
Standard IP access list 10
10 permit host 192.168.1.34 (0 matches)
20 permit host 192.168.1.50 (5 matches)
Extended IP access list 101
10 permit tcp host 192.168.1.33 any eq telnet (0 matches)
20 permit tcp host 192.168.1.49 any eq telnet (0 matches)
```

Based on the output, you can determine that RouterB is configured with two access lists: a standard ACL with an ID of 10 that permits traffic from the host with the IP address of 192.168.1.34 and the host with the IP address of 192.168.1.50, and an extended ACL with an ID of 101 that specifically permits Telnet traffic from the host with the IP address of 192.168.1.33 and the host with the IP address of 192.168.1.49. Issuing the show running-config | section line command on RouterB reveals that ACL 10, not ACL 101, is applied to the virtual terminal (vty) configuration on RouterB, as shown in the following output:

```
RouterB#show running-config | section line
line con 0
line aux 0
line vty 0 4
access-class 10 in
login
password cisco
```

In the output above, the access-class 10 in command configures vty lines from 0 through 4 to apply ACL 10 in the inbound direction. The standard ACL with an ID of 10 permits traffic with a source IP address that matches its own Serial 0 interface. Because of the implicit deny rule that applies to all ACLs, traffic from any other source is denied. Therefore, RouterA is unable to telnet to RouterB; ACL 10 on RouterB does not permit IP address.

ACLs are constructed by using the following syntax:

Standard: access-list access-list-number [permit | deny] [host | source source-wildcard | any]
Extended: access-list access-list-number [dynamic dynamic-name [timeout minutes]] {deny | permit} protocol source source-wildcard [operator [port]] destination destination-wildcard [operator [port]] [established] [precedence precedence] [tos tos] [log | log-input] [time-range time-range-name]
ACL 101 on RouterB does permit Telnet traffic. However, ACL 101 is not applied to the vty lines on RouterB. There are two ways to solve the problem in this scenario: by replacing the inbound ACL in RouterB's vty configuration with the extended ACL that has an ID of 101 or by reconfiguring ACL 10 with a source network or host IP address that will be matched when the source of the traffic is RouterA.

There are no outbound ACLs applied to RouterA's configuration. RouterA is configured with a single standard ACL that permits all traffic from the 192.168.1.0/24 network.

Original Exam B Question 58

You administer the network shown in the exhibit. You issue several extended ACLs on RouterA's Gi0/0 interface, including ACL 106, which permits IP traffic to destination 172.16.1.0/24. Multiple ip access-group commands are applied inbound to the same interface.

Which of the following statements is true regarding this configuration? (Select the best answer.)

- A. All traffic destined for the 172.16.1.0/24 network will be allowed.
- B. HostA will not be able to access the Internet.
- C. All IP traffic will be blocked on the Gi0/0 interface.
- D. A range of common threats will be mitigated.
- E. Only broadcast traffic will be allowed on the 172.16.1.0/24 network.

Correct Answer: A

Explanation

Based on the configuration shown in this scenario, all traffic destined for the 172.16.1.0/24 network will be allowed. Access control lists (ACLs) are used to control packet flow into and out of a router. They can either permit or deny packets based on source network address, destination network address, protocol, or port. Multiple ACLs can be used to accomplish more complex packet flow throughout a network. You have created multiple ACLs in this scenario, but they all use Internet Protocol (IP). When multiple ACLs that use the same protocol are applied to an interface, only the last ACL applied to the interface will affect traffic on the interface. The last ACL that is applied to interface Gi0/0 is ACL 106, which contains the statement `access-list 106 permit ip any 172.16.1.0 0.0.0.255`. This statement permits IP traffic from any source address to be sent to the 172.16.1.0/24 network. Any IP traffic that is destined for the 172.16.1.0/24 network and that is sent through the Gi0/0 interface will be allowed by this configuration.

A range of common threats will not be mitigated by the ACL in this scenario. Although the individual statements in this scenario are intended to mitigate common threats, such as IP spoofing, they should be applied as a single ACL rather than as separate ACLs.

The ACL configuration shown in this scenario will not block all IP traffic on Gi0/0. Because ACL 106 was applied to the interface last and ACL 106 permits IP traffic from any source address to be forwarded to the 172.16.1.0/24 network, no IP traffic that is destined for the 172.16.1.0/24 network will be blocked on Gi0/0. However, because no other IP traffic is explicitly permitted by ACL 106, all inbound IP traffic that is not destined for the 172.16.1.0/24 network will be denied by the implicit deny statement at the end of ACL 106.

HostA's Internet access will not be affected by the configuration in this scenario. The ACLs are applied to the Internet-connected interface, and HostA is connected to the internal network.

The configuration in this scenario will not permit only broadcast traffic on the 172.16.1.0/24 network. All IP traffic directed toward the 172.16.1.0/24 network will be permitted in this scenario.

Day 36 - CDP & LLDP

Included Exam B items: Q29, Q59

Original Exam B Question 29

You issue the following command on your router:

```
RouterA#show cdp neighbors
```

Which of the following information will be displayed? (Select 4 choices.)

- A. the software version running on the neighboring device
- B. the device ID of the neighboring device
- C. the IP address of the neighboring device
- D. the interface on RouterA that is connected to the neighboring device
- E. the interface on the neighboring device that is connected to RouterA
- F. the capabilities and product number of the neighboring device

Correct Answer: B, D, E, F

Explanation

The following information will be displayed when the show cdp neighbors command is issued:

- The device ID of the neighboring device
- The capabilities of the neighboring device
- The product number of the neighboring device
- The holdtime
- The local interface
- The remote interface

The following is sample output from the show cdp neighbors command:

```
RouterA#show cdp neighbors
```

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge

S - Switch, H - Host, I - IGMP, r - Repeater

Device ID Local Intrfce Holdtme Capability Platform Port ID

```
boson-1 Gig 0/1 117 sI WS-C2960-2 Gig 0/1
```

```
boson-2 Gig 143 sI WS-C2960-4 Gig 0/1
```

```
boson-3 Fas 0/3 102 RST 1841 Fas 0/1
```

```
boson-4 Fas 0/4 133 RSI 1841 Fas 0/2
```

The Device ID field indicates the host name, Media Access Control (MAC) address, or serial number of the neighboring device. The Local Intrfce field indicates the interface on the local device. The Holdtme field indicates the amount of time remaining before the Cisco Discovery Protocol (CDP) advertisement is discarded. The Capability field indicates the type of device:

e R-router

S-switch

e I-Internet Group Management Protocol (IGMP) device

e H-host

T-transparent bridge

e B-source-routing bridge

• r-repeater

The Platform field indicates the product number of the neighboring device. The Port ID field indicates the connected interface on the neighboring device.

You cannot discover the Internet Protocol (IP) address of a neighboring device or the software version running on a neighboring device by issuing the show cdp neighbors command. However, this information is included in the output of the show cdp neighbors detail command. The following is sample output from the show cdp neighbors detail command:

```
RouterA#show cdp neighbors detail
```

Device ID: boson-1

Entry address(es):

IP address: 10.1.1.2.3

Platform: WS-C2960-24TT-L, Capabilities: Switch IGMP

Interface: GigabitEthernet0/1, Port ID (outgoing port): GigabitEthernet0/1e

Holdtime: 117 sec

Version:

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 12.2(25)SEE2, RELEASE SOFTWARE (fc1)

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Compiled Tue 08-Nov-06 15:20 by smithc

advertisement version: 2

VTP Management Domain: ·boson*

Native VLAN: 1

Duplex: full

Management address(es):

Original Exam B Question 59

Which of the following commands should you issue to disable CDP globally on a router or switch? (Select the best answer.)

- A. cdp disable
- B. cdp stop
- C. no cdp enable
- D. no cdp run

Correct Answer: D

Explanation

You should issue the no cdp run command from global configuration mode to disable Cisco Discovery Protocol (CDP) globally on a router. CDP, which is enabled by default, is used to obtain information about neighboring devices. Disabling CDP globally prevents a router or switch from sending or receiving CDP advertisements on any interface. To re-enable CDP on a router or switch, you should issue the edp run command from global configuration mode.

The no cdp enable command does not disable CDP globally. You should issue the no cdp enable command from interface configuration mode to disable CDP on a single interface. The no cdp enable command is particularly helpful for situations in which you want to prevent a router or switch from sending sensitive device information over a link that is connected to a public network, such as the Internet, but you want to allow the router to continue sending and receiving CDP information over other interfaces. CDP is enabled by default on all interfaces except for asynchronous serial interfaces. To re-enable CDP on an interface, you should issue the edp enable command from interface configuration mode.

The edp stop and cdp disable commands include incorrect syntax and are not valid Cisco commands.

Day 37 - NTP

Included Exam B items: Q43, Q66

Original Exam B Question 43

Which of the following is enabled on a Cisco router when you issue the `ntp server` command from global configuration mode? (Select the best answer.)

- A. static client mode
- B. authentication
- C. symmetric active mode
- D. server mode
- E. broadcast client mode

Correct Answer: A

Explanation

Network Time Protocol (NTP) static client mode is enabled on a Cisco router when you issue the `ntp server` command from global configuration mode. NTP is used to synchronize the time on network devices. An NTP static client receives its time from an NTP server. The syntax of the `ntp server` command is `ntp server ip-address`, where `ip-address` is the Internet Protocol (IP) address of the NTP server that the client will use to receive its time.

NTP broadcast client mode is enabled on a Cisco router when you issue the `ntp broadcast client` command from interface configuration mode. An NTP broadcast client listens on the configured interface for NTP broadcasts from an NTP server, which the NTP client uses to adjust its time. The difference between a broadcast client and a static client is that a broadcast client can receive its time from any NTP server. By contrast, a static client receives its time from the NTP server specified in the `ntp server` command.

NTP authentication is enabled on a Cisco router when you issue the `ntp authenticate` command from global configuration mode. Authentication can be used with NTP to provide source verification for NTP synchronization. NTP authentication supports only Message Digest 5 (MD5) keys. To enable authentication on an NTP client, you should issue the following command set:

```
ntp authenticate
ntp authentication-key key-number md5 key
ntp trusted-key key-number
ntp server ip-address key key-number
```

To enable authentication on an NTP server, you should issue the following command set:

```
ntp authenticate
ntp authentication-key key-number md5 key
```

NTP server mode is enabled on a Cisco router when you issue the `ntp master` command from global configuration mode. The syntax of the `ntp master` command is `ntp master [stratum]`, where `stratum` is an NTP stratum value from 1 through 15; if the stratum value is not specified, the NTP server uses the default stratum value of 8. NTP servers not only synchronize time with NTP clients but also synchronize time with each other. Devices with higher stratum numbers receive time from devices with lower stratum numbers. For example, a stratum 2 device typically receives its time from a stratum 1 device, a stratum 3 device typically receives its time from a stratum 2 device, and so on.

NTP symmetric active mode is enabled on a Cisco router when you issue the `ntp peer` command from global configuration mode. A device in symmetric active mode attempts to mutually synchronize with another NTP host; the host might synchronize the peer, or it might be synchronized by the peer. The syntax of the `ntp peer` command is `ntp peer ip-address`, where `ip-address` is the IP address of the NTP host.

Original Exam B Question 66

You issue the following block of commands on RouterB:

```
RouterB(config)#ntp server 71.40.128.157
RouterB(config)#ntp server 128.227.205.3
RouterB(config)#ntp server 184.22.97.162
```

You want to determine from which of the three servers RouterB is currently synchronizing time. Which of the following commands would not achieve your goal? (Select the best answer.)

- A. showclock detail
- O 8B. show ntp associations detail
- C. show ntp associations
- D. show ntp status

Correct Answer: A

Explanation

Of the available choices, only the show clock detail command would not enable you to determine from which of the three Network Time Protocol (NTP) servers RouterB is synchronizing time. The show clock detail command displays the date and time as it is configured on the device and general information about the source of the configuration. However, this command does not reveal the Internet Protocol (IP) address or NTP peer status of an NTP source. The following is sample output from the show clock detail command:

```
RouterB#show clock detail
```

```
09:12:20.299 UTC Sat Jul 4 2015
```

```
Time source is NTP
```

Both the show ntp associations command and the show ntp associations detail command would enable you to determine from which of the three NTP servers RouterB is synchronizing time. The show ntp associations command displays both the address of the NTP server from which the client obtains its time and the address of the reference clock to which the NTP server is synchronized. When you issue show ntp associations with the detail keyword, you can additionally determine the IP address of the NTP peer from which time was synchronized, the NTP source authentication status, the NTP hierarchical status of the server from which time was obtained, whether the NTP peer passes basic sanity checks, whether NTP believes the time is valid, and the stratum of the NTP peer. The following is sample output from both the show ntp associations command and the show ntp associations detail command:

```
RouterB#show ntp associations
```

```
address ref clock st when poll reach delay offset disp
```

```
¥w128.227.205.3 ~GPS. 1 17 64 377 8.080 8.008 98.238
```

```
~71.40.128.157 =204.9.54.119 2 18 64 377 8.000 -321 1.816
```

```
~184.22.97.162 132.163.4.101 2 s 64 377 6.800 -314 1.134
```

```
* master (syncd), # master (unsyncd), + selected, - candidate, ~ configured
```

```
RouterB#show ntp associations detail
```

```
128.227.205.3 configured, our _master, sane, valid, stratum 2
```

```
ref ID .GPS., time BF6C06E0.55040FDS (10:30:59.212 UTC Sat Jul 4 2015)
```

```
our mode client, peer mode server, our poll intvl 64, peer poll intvl 64
```

```
root delay 0.00 msec, root disp 0.03, reach 1, sync dist 15875.977
```

```
delay 1.88 msec, offset -0.3598 msec, dispersion 15875.02
```

```
precision 2**16, version 3
```

```
org time BF6C06E1.0EES6FA3 (10:30:59.162 UTC Sat Jul 4 2015)
```

```
rcv time BF6CQ6E1.0F3ABCAS (16:30:59.045 UTC Sat Jul 4 2015)
```

```
xmt time BF6CQ6E1.0EBC84F9 (10:30:59.397 UTC Sat Jul 4 2015)
```

```
filtdelay = 1.88 8.00 8.00 8.00 8.00 8.00 8.00 8.00
```

```
filtoffset = -0.36 0.00 8.00 6.80 8.08 6.00 8.08 8.08
```

```
filtererror = 8.02 16000.0 16000.0 16000.0 16000.0 16000.0 16000.0
```

The presence of our_master in the output of the show ntp associations detail command indicates the status of the device at the NTP peer IP address of 128.227.205.3. Similarly, the asterisk (*) in the output of the show ntp associations command indicates that RouterB's NTP master is the device with the IP address of 128.227.205.3.

The show ntp status command would enable you to determine from which of the three NTP servers RouterB is synchronizing time. The show ntp status command displays no information when NTP is not running on a device. When NTP is running, the show ntp status command provides information about whether the local clock is synchronized, the local clock's stratum level, and the IP address of the NTP peer that the local device is using as a reference clock. The following is sample output from the show ntp status command:

```
RouterB#show ntp status
```

```
Clock is synchronized, stratum 2, reference is 128.227.205.3
```

```
nominal freq is 250.0000 Hz, actual freq is Hz, precision is 2**16
```

reference time is BF6BF651.8FE73277 (10:25:36.862 UTC Sat Jul 4 2015)
clock offset is -0.0794 msec, root delay is 1.85 msec
root dispersion is 0.15 msec, peer dispersion is 0.03 msec

Day 38 - DNS

Included Exam B items: Q3, Q64

Original Exam B Question 3

You administer the example.com network in the topology. Router1 is configured as a DNS server. The www server is an HTTP server that is used by all departments. The ftp server is an FTP server that is used only by the Web department. The user at the Accounting workstation reports that neither the Accounting department nor the Sales department can access the www server by entering the server's host name or FQDN in a web browser. Attempts to access the www server by entering the server's IP address in a web browser also fail. The Web Admin user reports that the www server can be accessed from the Web department only by entering the server's IP address in a web browser. You want to repair the network configuration so that all workstations can ping the www server by IP address, host name, and FQDN. In addition, you want all the workstations that are connected to Switch2 to be able to access the www server by using HTTP. Finally, you want to ensure that only users from the Web department can access the ftp server by using passive FTP. Access the console of any device by clicking the device in the topology, and repair the configurations. You should make no changes other than the ones required to accomplish the task. You should not add more ACL statements to the existing configuration. In addition, your changes should not modify the sequencing or identifiers of any existing ACLs.

Source result: NetSim lab task - no separate answer/explanation is included in the uploaded Exam B PDF.

Original Exam B Question 64

Which of the following is most likely a symptom of a DNS problem? (Select the best answer.)

- A. an error message indicating that the host example.com could not be found
- B. a message indicating that an attempt to connect to 192.168.1.137 timed out
- C. workstation's IP address assignment of 169.254.0.17
- D. transfer speeds from a remote host that are significantly slower than normal

Correct Answer: A

Explanation

Of the available choices, an error message indicating that the host example.com could not be found is most likely a symptom of a Domain Name System (DNS) problem. DNS is a hierarchical database that provides a cross-reference between a fully qualified domain name (FQDN) and the corresponding Internet Protocol (IP) address of a computer on a network. An FQDN includes the host name of a device and the domain name to which that device is connected. For example, DNS enables you to access a computer named server1 on the boson.com domain by using the FQDN server1.boson.com instead of the IP address that is assigned to server1. When a host receives a response from a DNS server, the host creates an entry in its cache for the host name or FQDN and the corresponding IP address. As long as a cache entry exists for a host or FQDN, no further queries will be sent to the DNS server for that particular host or FQDN. Cache entries greatly reduce the amount of time required to resolve a host name or FQDN to its corresponding IP address. DNS must be functional in order to connect to another host by using a host name or an FQDN. However, name resolution failures do not necessarily indicate that DNS is nonfunctional. For example, if an organization subscribes to a DNS-based content-filtering service, users will likely encounter name resolution failures when they attempt to connect to sites that are filtered by the service. When a DNS-based content-filtering service is used, the DNS servers hosted by the service are configured as the upstream DNS resolvers for the organization. This configuration ensures that local host names are always resolved locally and that the content-filtering service is used to resolve all external host names. A local workstation's IP address assignment of 169.254.0.17 is an indication of a Dynamic Host Configuration Protocol (DHCP) problem, not a DNS problem. DHCP can be used to automatically assign IP addresses to host computers. If DHCP is not available, a Windows computer will automatically assign an Automatic Private IP Addressing (APIPA) address. The range of IP addresses from 169.254.0.0 through 169.254.255.255 has been reserved for APIPA addresses by the Internet Assigned Numbers Authority (IANA), a nonprofit organization that is responsible for assigning IP addresses and managing root servers. A computer that has an APIPA address continually checks the

network for a DHCP server. When a DHCP server becomes available, the computer releases its APIPA address and leases an IP address from the DHCP server.

Transfer speeds from a remote host that are significantly slower than normal indicate a bandwidth problem, not a DNS problem. Internet service providers (ISPs) offer tiers of bandwidth to which users can subscribe. Each tier has a maximum bandwidth, which is the fastest rate at which a user can download data over the connection. One way to address slow transfer speeds is to upgrade bandwidth. However, slow transfer speeds can also be caused by background programs that are transferring information from the Internet without the user's knowledge. It is therefore important to use tools like the Windows netstat command to determine how bandwidth is being used at a given time.

An error message indicating that an attempt to connect to 192.168.1.137 timed out can be a symptom of a number of problems. However, it is not likely to be a symptom of a DNS problem in this scenario because the error message indicates a problem connecting to an IP address, not to a host name or an FQDN. An inability to connect to a given IP address can range from hardware issues to routing issues to a disconnected host on the remote end.

Day 39 - DHCP

Included Exam B items: Q23, Q36

Original Exam B Question 23

NetworkA: 10.10.2.0/24

NetworkB: 10.10.3.0/24

DHCP Server: 10.10.3.5

You are the administrator for the network shown above. DHCP services for the network are provided by the DHCP server on NetworkB. DHCP services are not running on the routers.

Which of the following commands should you issue to enable clients on NetworkA to receive IP addresses from the DHCP server? (Select the best answer.)

- A. RouterB(config-if)#ip helper-address 10.10.3.5
- B. RouterA(config-if)#ip helper-address 10.10.1.2
- C. RouterA(config-if)#ip helper-address 10.10.3.5
- D. RouterA(config-if)#ip helper-address 10.10.1.1
- E. RouterB(config-if)#ip helper-address 10.10.1.2
- F. RouterB(config-if)#ip helper-address 10.10.1.1

Correct Answer: C

Explanation

To allow clients on NetworkA to receive Internet Protocol (IP) addresses from the Dynamic Host Configuration Protocol (DHCP) server located on NetworkB, you should issue the ip helper-address 10.10.3.5 command on the Fa0/0 interface of RouterA. The ip helper-address command configures an interface to forward DHCP broadcasts across a network to the IP address of a DHCP server. The DHCP server will respond to the request and begin the DHCP discovery process.

By default, the ip helper-address command configures an interface to forward broadcasts to the following User Datagram Protocol (UDP) ports:

e UDP port 37 · Time Protocol

UDP port 49 · Terminal Access Controller Access-Control System (TACACS)

e UDP port 53 · Domain Name System (DNS)

e UDP port 67 · Bootstrap Protocol (BOOTP) and DHCP Server

e UDP port 68 · BOOTP and DHCP Client

e UDP port 69 · Trivial File Transfer Protocol (TFTP)

e UDP port 137 · Network Basic Input/Output System (NetBIOS) Name Service

e UDP port 138 · NetBIOS Datagram

You can issue the ip forward-protocol udp port command to configure the router to forward broadcasts for additional UDP ports.

The use of the ip helper-address command is not limited to a single unicast address. If a network contains multiple DHCP servers, an entry for each server could be configured on the router. The same would hold true for a network running DHCP, DNS, and TFTP on multiple servers with different IP addresses. For example, if DHCP requests were serviced at IP address 10.10.3.5, DNS at 10.10.3.4, and TFTP at 10.10.3.3, you would issue the following commands:

```
RouterA(config-if)#ip helper-address
```

```
RouterA(config-if)#ip helper-address 10.10.3.4
```

```
RouterA(config-if)#ip helper-address 10.10.3.5
```

These commands would be displayed in the output of the show running-config command as shown below:

```
!  
interface FastEthernet0/e  
ip address 10.10.2.1 255.255.255.0  
ip helper-address 10.10.3.3  
ip helper-address 10.10.3.4  
ip helper-address 10.10.3.5  
no ip directed-broadcast  
no shutdown  
!
```

UDP broadcasts are not forwarded by routers. Therefore, the `ip helper-address` command must be issued on interface Fa0/0 of RouterA for UDP broadcast traffic to reach NetworkB. Issuing the `ip helper-address` command on RouterB will have no effect on broadcast traffic emanating from NetworkA.

The addresses 10.10.1.1 and 10.10.1.2 are the interface IP addresses that connect RouterA and RouterB, and these routers do not provide DHCP services for either network. When you configure the `ip helper-address` command, you should use the IP address of the DHCP server or a server farm.

Original Exam B Question 36

RouterA has an internal interface of 10.1.1.1/24 and an external interface of 11.1.1.1/24. RouterA is configured as a DHCP server. You want to provide the DHCP client computers with the proper gateway information.

Which of the following commands should you issue from DHCP pool configuration mode? (Select the best answer.)

- A. `next-server 11.1.1.1`
- B. `dns-server 10.1.1.1`
- C. `default-router 11.1.1.1`
- D. `next-server 10.1.1.1`
- E. `default-router 10.1.1.1`
- F. `dns-server 11.1.1.1`

Correct Answer: E

Explanation

You should issue the `default-router 10.1.1.1` command from Dynamic Host Configuration Protocol (DHCP) pool configuration mode to provide the DHCP client computers with the proper gateway information. A default gateway is a router that is used to deliver packets that are destined for an address outside the local network. Typically, the default gateway for a host is the router interface that is local to the host. Router1's 10.1.1.1 interface is local to the client computers and would therefore be the default gateway for the client computers. You can specify up to eight gateway addresses with the `default-router` command. If multiple routers are specified, they should be listed in the order they should be used by the DHCP client. For example, if you were to issue the `default-router 10.1.1.1 10.1.1.2` command, then the router at 10.1.1.2 would be used by the client only if the router at 10.1.1.1 were unavailable.

You should not issue the `default-router 11.1.1.1` command. The Internet Protocol (IP) address 11.1.1.1 is configured for the external router interface, not the internal router interface; the gateway address should be the IP address of the interface that is local to the host computers.

Neither the `next-server 10.1.1.1` command nor the `next-server 11.1.1.1` command will configure your router to provide the DHCP client computers with the proper gateway information. The `next-server` command is used to specify the next server used in the boot process of a DHCP client. This server is often a Trivial File Transfer Protocol (TFTP) server. You can specify up to eight server addresses with the `next-server` command. If you specify multiple servers, they will be used by DHCP clients in a round-robin fashion; the first client will use the first address, the second client will use the second address, and so on.

Neither the `dns-server 10.1.1.1` command nor the `dns-server 11.1.1.1` command will configure your router to provide the DHCP client computers with the proper gateway information. The `dns-server` command is used to specify the Domain Name System (DNS) servers that should be used by a DHCP client. DNS is used to map host names to IP addresses. You can specify up to eight server addresses with the `dns-server` command. If multiple servers are specified, they should be listed in the order they should be used by the DHCP client. For example, if you were to issue the `dns-server 10.1.1.1 10.1.1.2` command, the DNS server at 10.1.1.2 would be used by the client only if the DNS server at 10.1.1.1 were unavailable.

Day 40 - SNMP

Included Exam B items: Q10

Original Exam B Question 10

Which of the following best describes the purpose of SNMP? (Select the best answer.)

- A. to create VPNs
- B. information about directly connected Cisco devices
- C. to transfer files
- D. to send email
- E. to manage network devices

Correct Answer: E

Explanation

Simple Network Management Protocol (SNMP) is used to manage network devices. SNMP can be used to remotely monitor and configure a wide variety of network devices, such as routers, switches, and network printers. SNMP uses User Datagram Protocol (UDP) ports 161 and 162. SNMP version 1 (SNMPv1) and SNMPv2 use community strings to provide authentication. However, neither SNMPv1 nor SNMPv2 uses encryption; all data and community strings are sent in clear text. A malicious user can sniff an SNMP community string and use it to access and modify network devices.

SNMPv3 is an enhancement to the SNMP protocol that uses encryption to provide confidentiality, integrity, and authentication. SNMPv3 is also the first version of SNMP to support user and group configuration in its authentication security model. However, before you can configure SNMPv3 users, you must configure the following:

• An SNMPv3 group that maps users to views

- The Internet Protocol (IP) address or port number for the remote SNMP agent where the user is configured
- The SNMP engine ID by issuing the `snmp-server engineID` command

Although you can configure SNMP to perform automated backups of configuration files by using Management Information Base (MIB) object IDs (OIDs), SNMP itself is not used to transfer files. SNMP relies upon other file transfer protocols, such as File Transfer Protocol (FTP), Trivial FTP (TFTP), and Secure FTP (SFTP).

SNMP can collect information from many different network devices, not just neighboring Cisco devices. By contrast, Cisco Discovery Protocol (CDP) is used to advertise and discover only directly connected Cisco devices.

SNMP is not used to send email. Simple Mail Transfer Protocol (SMTP) is used to send email. Post Office Protocol 3 (POP3) and Internet Message Access Protocol 4 (IMAP4) are used to receive email. SNMP is not used to create virtual private networks (VPNs). To create a VPN, you would typically use a protocol that can encrypt the data on the virtual network, such as IP Security (IPSec). A VPN is often used when it is necessary to connect two locations that are separated by a public network, such as the Internet.

Day 41 - Syslog

Included Exam B items: Q18

Original Exam B Question 18

Drag each Syslog message-logging keyword to its corresponding Syslog level.

> 3 3

Correct Answer: 0 - emergencies

1 - alerts

2 - critical

3 - errors

4 - warnings

5 - notifications

6 - informational

7 - debugging

Explanation

Cisco debug messages and log messages are divided into the following severity levels:

-
- e 1-alerts
- e 2-critical
- e 3-errors
- 4-warnings
- e 5 -notifications
- 7-debugging

You can filter log messages on the console by severity level by issuing the logging console severity-level command, or you can filter log messages to a Syslog server by issuing the logging trap severity-level command. When the logging console or logging trap command is issued with a severity-level parameter, messages with the specified severity level and all lower-numbered severity levels will be displayed or sent, respectively.

Messages are formatted in the Berkeley Software Distribution (BSD) Syslog format, which is a percent sign (%) followed by a facility code, a severity code, and a mnemonic code. The three codes are separated by dashes. Consider the following message:

·Dec 1 14:20:07.059: X%DUAL-5-NBRCHANGE: IP-EIGRP(0) 1: Neighbor 192.168.1.2 (FastEthernet0/0) is down: interface down

In this Syslog message, the facility code is DUAL, the severity code is 5, which is equivalent to notifications, and the mnemonic code is NBRCHANGE. The dash-separated code is followed by a colon and the human-readable text of the log message.

Emergencies and alerts indicate a severe hardware or software problem with the device. These messages need to be addressed immediately.

Critical, error, and warning messages indicate something that might impact the device. For example, interface up/down state changes are displayed as errors at level 3.

Notifications and informational messages are routine messages but still might indicate a problem. Route flaps, neighbor adjacencies, and interface protocol up/down transitions are displayed as notifications at level 5.

Debugging messages appear only as the result of issuing the debug command. After debugging, always remember to issue the no debug all command to stop collecting data.

Day 42 - SSH & Device Management

Included Exam B items: Q60, Q62

Original Exam B Question 60

You want to configure SSH for incoming vty connections on a new router. The router is running a K9 IOS image but has not yet been configured with a host name, a domain name, or an RSA key pair. In addition, the vty lines are not yet configured to accept incoming SSH connections.

You issue the `ip ssh time-out 60` command from global configuration mode to configure the router with a 60-second timeout. Which of the following messages will you most likely receive? (Select the best answer.)

- A. Invalid input detected at '*' marker.
- B. Please enable SSH as a transport mode.
- C. Please create RSA keys to enable SSH.
- D. Please define a hostname other than Router.
- E. Please define a domain-name first.

Correct Answer: C

Explanation

You will most likely receive the Please create RSA keys to enable SSH. message when you issue the `ip ssh time-out 60` command from global configuration mode. To enable Secure Shell (SSH) for virtual terminal (vty) lines on a Cisco router, you should complete the following steps:

1. Configure the router with a host name other than Router by issuing the `hostname` command.
2. Configure the router with a domain name by issuing the `ip domain-name` command.
3. Generate an RSA key pair for the router by issuing the `crypto key generate rsa` command.
4. Configure the vty lines to use SSH by issuing the `transport input ssh` command from line configuration mode.

SSH is often used as a secure replacement for Telnet to manage network devices. In order for SSH to be enabled on a Cisco device, the device must be running a K9 IOS image, which provides cryptographic functionality.

You will not receive the Invalid input detected at '*' marker. message when you issue the `ip ssh time-out 60` command in this scenario. You would receive the Invalid input detected at '*' marker. message if you were to mistype the time-out keyword or if you were to try to configure the SSH timeout with a value greater than 120 seconds. Although SSH is not yet enabled in this scenario, the router will accept the `ip ssh time-out 60` command as a valid configuration. The `ip ssh time-out 60` command would appear in the configuration if you were to issue the `show running-config` command.

You will not receive the Please define a hostname other than Router. message when you issue the `ip ssh time-out 60` command in this scenario. However, because you have not configured the router with a host name other than the default name of Router, you would receive the Please define a hostname other than Router. message if you were to issue the `crypto key generate rsa` command. To configure a router with a host name other than the default, you should issue the `hostname` command from global configuration mode.

You will not receive the Please define a domain-name first. message when you issue the `ip ssh time-out 60` command in this scenario. However, if you had configured the router with a valid host name but had not configured the router with a domain name, you would receive the Please define a domain-name first. message if you were to issue the `crypto key generate rsa` command. In this scenario, you have configured neither the domain name nor the host name. To configure a router with a domain name, you should issue the `ip domain-name` command from global configuration mode.

You will not receive the Please enable SSH as a transport mode. message when you issue the `ip ssh time-out 60` command in this scenario. The Please enable SSH as a transport mode. message is not a warning message that is displayed on Cisco routers. You can issue the `transport input ssh` command to configure SSH as the transport mode for vty lines.

Original Exam B Question 62

Refer to the following command output from SwitchA:
`SwitchA#show running-config | begin line`

```
line con 0
line aux 0
line vty 0 15
```

Based on SwitchA's configuration, in which of the following ways can users not manage SwitchA? (Select the best answer.)

- A. the AUXport
- B. SSH
- C. the console port
- D. Telnet

Correct Answer: A

Explanation

Based on SwitchA's configuration, users cannot manage SwitchA by using the auxiliary (AUX) port in this scenario. The login command is enabled by default for the AUX line. Therefore, you will not see the login command in the running configuration. However, no password has been configured for the AUX port. If the login command is issued and a password has not been configured, the device will be inaccessible through the point of access until a password has been configured.

Users can manage SwitchA by using the console port, Secure Shell (SSH), or Telnet. To determine the ways in which users can manage SwitchA, you should issue the show running-config | begin line command from privileged EXEC mode, as shown in the following output:

```
SwitchA#show running-config | begin line
```

```
line con 0
login local
line aux 0
line vty 0 15
login local
transport input telnet ssh
<output omitted>
```

Any form of administrative access to a network device should always be authenticated. The simplest authentication mechanism is a password assigned to a particular point of access. You can issue the password command to specify a password for the console, the auxiliary (AUX) line, or a particular virtual terminal (vty) line. Once a password has been configured, you can then issue the login command to specify that a user must authenticate before access to the device is granted.

However, in this scenario, the login local command has been issued for both the console port and the vty lines. The login local command configures the specified line to use the device's local user database for authentication. If the login local command is specified, it is not necessary to issue the password command in line configuration mode. Instead, you should ensure that an appropriate user name and password have been configured in global configuration mode.

In addition to the login local command, the transport input telnet ssh command has been issued for the vty lines on SwitchA. The transport input command enables an administrator to limit the shell protocols that can be accessed on the lines. In this scenario, the vty lines can be accessed by using either Telnet or SSH.

It is important to note that SSH requires additional configuration on a Cisco device before SSH connections can be supported on vty lines. To correctly configure an SSH server on a Cisco device, you should ensure that the device is configured with a host name, a domain name, and an RSA encryption key. You can configure the host name by issuing the hostname host-name command from global configuration mode. You can configure a domain name by issuing the ip domain name domain-name command from global configuration mode. Finally, you can configure the RSA encryption key by issuing the crypto key generate rsa command from global configuration mode.

Days 44-45 - NAT

Included Exam B items: Q1

Original Exam B Question 1

You administer the network depicted in the topology.

You want to configure static NAT on Router1 so that the DMZ network's Intranet host, which has been assigned the inside local

address of 198.51.100.10, can communicate with hosts on the Internet by using the inside global address of 203.0.113.4. In addition, you want to configure PAT on Router1 so that every inside local address on the LAN network uses the inside global address that has been assigned to Router1's Serial 0/0 interface to communicate with hosts on the Internet.

Any ACLs you configure on the devices should use an identifier of 10. You should not configure any routing protocols or static routes anywhere in the topology.

When you have completed your task, the Intranet host on the DMZ network should be able to ping the Web Server's 192.0.2.14

IP address. Similarly, any host on the LAN network should be able to ping the Web Server's 192.0.2.14 IP address.

Click the devices in the topology to access their consoles and complete the configuration.

Web Server Sales-PC

Source result: NetSim lab task - no separate answer/explanation is included in the uploaded Exam B PDF.

Days 46-47 - QoS

Included Exam B items: Q37

Original Exam B Question 37

Which of the following four types of traffic should you place in a priority queue? (Select the best answer.)

- A. video traffic
- B. webtraffic
- C. VoIPtraffic
- D. database traffic

Correct Answer: C

Explanation

Of the choices provided, you should place Voice over Internet Protocol (VoIP) traffic into a Quality of Service (QoS) priority queue. VoIP traffic is very sensitive to latency, jitter, and packet loss, which can cause clarity problems or dropped calls. A priority queue will ensure that higher-priority traffic, such as VoIP, will always be sent before lower-priority traffic.

Video traffic is also sensitive to delay and jitter. However, video traffic is less sensitive than VoIP traffic. To prevent video traffic from consuming all of a network's bandwidth, you might consider implementing traffic policing. Traffic policing enforces strict bandwidth limits whereby traffic that exceeds the committed information rate (CIR) is dropped. Although policing can cause dropped traffic, policing is preferable to traffic shaping, which can cause stuttering issues for video traffic. This happens because shaping buffers bursty traffic, such as video.

Database traffic is not very sensitive to delay or jitter, but it is extremely sensitive to packet loss. Therefore, traffic policing is not a good choice for database traffic. Instead, shaping should be used for database traffic. Traffic shaping buffers, or stores, traffic that exceeds a maximum rate. When bandwidth becomes available, the buffered traffic is sent with no packet loss.

Web traffic is not sensitive to delay, jitter, or packet loss; packets that are lost can simply be retransmitted. As a result, web traffic is often placed in a best-effort queue, in which traffic is sent whenever spare bandwidth is available.

Day 48 - Security Fundamentals

Included Exam B items: Q15, Q17, Q28, Q44, Q65, Q72

Original Exam B Question 15

Which of the following combinations represents a single-factor authentication method? (Select the best answer.)

- A. a fingerprint, a retina scan, and a password
- B. a smart card, a password, and a PIN
- C. a password and a PIN
- D. a password, a fingerprint, and a smart card

Correct Answer: C

Explanation

Of the available options, the combination of a password and a personal identification number (PIN) represents a single-factor authentication method. A single-factor authentication method refers to the use of only one of the three common methods to verify a user's identity. The three authentication factors are something you know, something you have, and something you are. A password and a PIN are knowledge factor access control methods, which are pieces of information that you know. Because a password and a PIN are both something you know, when the two are used in combination with each other they represent a single-factor authentication method.

Two-factor, or dual-factor, authentication refers to the use of two different factors to authenticate a user. For example, a password and a code that is automatically generated at the time of authentication by a device or token is two-factor authentication because it requires both knowledge of the password and possession of the device.

The combination of a fingerprint, a retina scan, and a password is also an example of a two-factor authentication method. The fingerprint and retina scanners are both biometric, which are inherent factor access control methods, and the password is something the user knows. An inherent factor is an attribute that is part of the user, such as fingerprints, retinal patterns, and voice patterns. Biometric scanners are used to obtain a scan of a user's physical attributes and are considered the most effective method of securing physical access to network infrastructure. An authentication system compares the scan to a previously stored scan; if the scans match, the user will be authenticated.

The combination of a smart card, a password, and a PIN is also a two-factor authentication method. A smart card is a physical item a user possesses, which is an ownership factor access control method, and a password and a PIN are both something the user knows. A smart card contains a memory chip that is encoded with a unique digital identifier. The digital code stored on a smart card is read by a smart card reader and compared to the list of authorized users. If the digital signature matches a user on the list, the user will be authenticated.

Multifactor authentication refers to the use of two or three factors to authenticate a user. Therefore, dual-factor authentication is also sometimes known as multifactor authentication.

Original Exam B Question 17

You issue the following commands on RouterA:

```
no service password-encryption
username admin password Z80$0n!A
```

Which of the following are true? (Select 2 choices.)

- A. The password has been issued with type 7 encryption.
- B. The password will be stored with type 0 encryption.
- C. The password has been issued with type 5 encryption.
- D. The password has been issued with type 0 encryption.
- E. The password will be stored with type 7 encryption.
- F. The password will be stored with type 5 encryption.

Correct Answer: B, D

Explanation

The password has been issued with type 0 encryption and will be stored with type 0 encryption in the running configuration. Type 0 means that the text of the password has been issued and stored as plain text. Therefore, the password is visible to anyone with access to the router's running

configuration. The `no service password-encryption` command that is issued in this scenario disables the encryption of future passwords in the running configuration if encryption was enabled. By default, the password encryption service is not enabled on Cisco devices.

The `username user-name password [encryption-type] password` command creates a user with the name specified by `user-name` and the password specified by `password`. There are two encryption-type values available for this command: 0 or 7. Issuing the `username admin password Z80$0n!A` command in this scenario creates a user named `admin` with a plain text password of `Z80$0n!A`. Therefore, issuing the `username admin password Z80$0n!A` command is the same as issuing the `username admin password 0 Z80$0n!A` command.

The password has not been issued with type 7 encryption and will not be stored with type 7 encryption. Type 7 encryption is performed by a Cisco algorithm. This reversible encryption type is considered weak. To configure a password with type 7 encryption, you should issue the `username user-name password 7 password` command, where `password` is a type 7-encrypted version of the password you want to store. Type 7 encryption is also used by the `service password-encryption` command to encrypt all current and future passwords in the running configuration.

The password has not been issued with type 5 encryption and will not be stored with type 5 encryption. Type 5 encryption is the term Cisco uses for Message Digest 5 (MD5) hashes of passwords. The `username user-name password [encryption-type] password` command does not support this type of encryption. However, the `username user-name secret [encryption-type] password` command can be used to both issue and store a type 5 password for a given user name. For example, issuing the `username user-name secret 0 Z80$0n!A` command will issue the password in plain-text format but store it in the running configuration as a type 5 password.

The `enable secret password` command also stores passwords using type 5 encryption. Although typically used to create a type 0 password, the `enable password {password | [encryption-type] encrypted-password}` command can also be used to store a type 5 password. However, you must already have the password in MD5 hashed format. The `enable secret password` command generates and stores the MD5 hash for you.

Original Exam B Question 28

Which of the following commands could you use to disable authentication for vty, console, or AUX connections to a router? (Select the best answer.)

- A. `login none local`
- B. `aaa authentication login disable none`
- C. `aaa authentication login local none`
- D. `aaa authentication login default local`

Correct Answer: B

Explanation

You could use the `aaa authentication login disable none` command to disable authentication for console, virtual terminal (vty), or auxiliary (AUX) connections to a router. The `aaa authentication login` command specifies the authentication methods that are supported when a user logs in to the system. The syntax for the `aaa authentication login` command is `aaa authentication login {default | list-name} method? [method2...]`, where `list-name` is an identifier for a list of authentication methods and `method7` is at least one of the following authentication methods:

Authentication Method | Description

`enable` uses the `enable password`

`krb5` uses Kerberos 5

`krb5-telnet` uses Kerberos 5 for Telnet

`line` uses the line password

`local` uses the local user database

`local-case` uses the case-sensitive local user database

`none` uses no authentication

`group radius` uses a group of Remote Authentication Dial-In User Service (RADIUS) servers

`group tacacs+` uses a group of Terminal Access Controller Access Control System Plus (TACACS+) servers

`group group-name` uses a subset of RADIUS or TACACS+ servers named `group-name`

Therefore, the `aaa authentication login disable none` command defines an authentication list named `disable` that uses no authentication. After the named list is created, you can apply the authentication list to router access methods. For example, if you want to apply the list to vty ports, you should issue the following commands:

```
Router(config)#line vty 0 4
```

```
Router(config-line)#login authentication disable
```

After you issue these commands, any future attempts to log in to the vty ports would not require authentication, because the named list uses no authentication. It is important to note that the `login authentication disable` command does not disable authentication; it applies an authentication list named `disable` to the vty ports.

In order to use AAA authentication with an authentication server, you must ensure that an authentication server is available on the network. To prepare for the possibility that the authentication server might become unavailable, you should configure a router so that it will use another method, such as the local user database or the enable password, in the absence of the authentication server. For example, the `aaa authentication login default group radius enable` command uses a RADIUS server for authentication; if the RADIUS server is unavailable, the enable password should be used. The `aaa authentication login default local` command would apply local user database authentication to all means of access that do not have a named authentication list specified.

Consequently, if you have not applied an authentication method for console, AUX, or vty connections, this command would force those connections to use the local user database for authentication.

The `aaa authentication login local none` command and the `aaa authentication login none local` command are not valid because they use invalid authentication list names. The following keywords cannot be used as an authentication list name:

- `auth-guest`
- `enable`
- `guest`
- `if-authenticated`
- `if-needed`
- `krb5`
- `krb-instance`
- `e krb-telnet`
- `e line`
- `local`
- `none`
- `radius`
- `remd`
- `e tacacs`
- `tacacsplus`

Original Exam B Question 44

Which of the following best describes authentication? (Select the best answer.)

- A. the process of establishing a user's accounts upon hire
- B. the process of verifying a user's identity
- C. the process of verifying the level of access configured for a user
- D. the process of recording the use of resources

Correct Answer: B

Explanation

Authentication is the process of verifying a user's identity. The following list defines the three phases of the Authentication, Authorization, and Accounting (AAA) process:

- Authentication - the process of verifying a user's identity
- Authorization - the process of verifying the level of access configured for a user
- Accounting - the process of recording the use of resources

AAA systems manage user activity. AAA systems are typically more sophisticated than simple password authentication systems, such as a local password database. Two common AAA systems are Remote Authentication Dial-In User Service (RADIUS) and Terminal Access Controller Access-Control System Plus (TACACS+).

RADIUS is a standard AAA protocol created by the Internet Engineering Task Force (IETF). Compared

to TACACS+, RADIUS has several limitations. For example, RADIUS encrypts only the password in Access-Request packets; it does not encrypt the entire contents of the packet like TACACS+ does. TACACS+ is a Cisco-proprietary protocol used during AAA operations. TACACS+ provides more granular and flexible control over user access privileges. For example, the AAA operations are separated by TACACS+, whereas RADIUS combines the authentication and authorization services into a single function. Because TACACS+ separates these functions, administrators have more control over access to configuration commands. In addition, TACACS+ encrypts the entire contents of packets, thus providing additional security.

Original Exam B Question 65

Drag each basic security program element to its description. An element can be used only once. Use all elements.

Correct Answer: User awareness - provides employees with information about protecting confidential data
User training - familiarizes users with corporate security policies in a formal and mandatory fashion
Physical access control - protects infrastructure locations such as network closets

Explanation

User awareness is the security program element that provides employees with information about protecting confidential data. User training is the security program element that familiarizes users with corporate security policies in a formal and mandatory fashion. Physical access control is the security program element that protects infrastructure locations such as network closets.

Educating users about enterprise security is an important step in ensuring the confidentiality and integrity of data. Through education, users without a technological background can learn to identify and thus avoid falling prey to common attacks. For example, users who have been educated on the dangers of phishing emails, pharming, and social engineering attacks will be less likely to click malicious links in emails than those who have not been educated on such tactics. Phishing is an attempt to compromise user information by disguising a malicious email as a legitimate email. Pharming is the use of a legitimate service to redirect users to a malicious or compromised site. Social engineering is the simple exploitation of human trust and instinctive behavior. The goal of both user awareness and user training is to minimize the likelihood of users becoming victims of their own instincts.

User awareness is different from user training in that user training is a more formal and mandatory process. Making users aware of potential threats might involve little more than sending emails or making announcements that describe the threat. User training, on the other hand, is formally instructing users by using a published company security policy that outlines how to identify a given threat and how to respond to it. An example of user training might be guiding users through a company's password complexity policy. In this case, users would be taught how to create secure passwords in order to mitigate common password attacks, such as a brute-force attack. A brute-force attack occurs when an attacker uses every possible combination of characters in an attempt to guess a password.

Although not specific to users, physical access control allows administrators to protect sensitive equipment and data from accidental or malicious compromise by users. For example, installing a proximity badge reader on a network closet door prevents users, visitors, or malicious actors from accessing, corrupting, or stealing the hardware that keeps the local area network (LAN) running. Physical access controls can be as simple as a padlock or as elaborate as a mantrap, which is a system that is intended to prevent unauthorized people from following an authorized person into a secured location. A mantrap, or access control vestibule, consists of a set of two locked doors that operate in a manner that prevents both doors from being opened at the same time. People who enter the mantrap must be identified before they are allowed to exit the mantrap into the secure area. Security guards can use facial recognition to verify the identity of proximity badge holders.

Original Exam B Question 72

You are implementing common Layer 2 security measures on a Cisco switch. You create a new VLAN with an ID of 4. No devices operate on VLAN 4. You issue the following commands on a switch interface:

```
switchport access vlan 4  
switchport mode access
```

Which of the following Layer 2 security measures are you implementing? (Select 2 choices.)

- A. configuring the port mode manually
- B. enabling port security on an access port
- C. disabling an unused port
- D. moving the port to an unused VLAN
- E. disabling DTP on a port

Correct Answer: A, D

Explanation

You are moving the port to an unused virtual local area network (VLAN) by issuing the switchport access vlan 4 command in this scenario. In addition, you are configuring the port mode manually by issuing the switchport mode access command. By default, every network interface on a Cisco switch is an active port. Before you deploy a switch on a network, you should take steps to ensure that every trunk port and access port on the switch is secured and that every unused port on the switch is disabled.

Moving an unused port to an unused VLAN creates a logical barrier that prevents rogue devices from communicating on the network should such a device connect to the port. To move an access port to an unused VLAN, you should issue the switchport access vlan vian-id command on the port, where vlan-id is the ID of the unused VLAN. When you move an unused port to an unused VLAN, you should also manually configure the port as an access port by issuing the switchport mode access command and shut down the port by issuing the shutdown command.

To manually configure an access port, you should issue the switchport mode access command in interface configuration mode. To manually configure a trunk port, you should first issue the switchport trunk encapsulation protocol command in interface configuration mode, where protocol is the trunk encapsulation protocol you want to use, and then issue the switchport mode trunk command in interface configuration mode.

You are not disabling Dynamic Trunking Protocol (DTP) on a port if you issue the commands in this scenario. By default, all interfaces on a Cisco switch will use DTP to automatically negotiate whether an interface should be a trunk port or an access port. The transmission of DTP packets over an interface can be exploited by a malicious user to obtain information about the network or to convert an interface that should be an access port into a trunked port. You should issue the switchport nonegotiate command on a manually configured port to prevent any attempts by the switch to negotiate by using DTP. Manually configuring interfaces to use either trunk mode or access mode effectively disables DTP and ensures that the traffic on those ports is restricted to the intended purpose. Even so, you should issue the switchport nonegotiate command on a manually configured trunk port to prevent any attempts by the switch to negotiate by using DTP, because a manually configured trunk port will continue to send DTP frames.

You are not disabling an unused port by issuing the commands in this scenario. Disabling an unused port creates a barrier that prevents rogue devices from communicating on the network should such a device connect to the port. To disable an unused port on a switch, you should issue the shutdown command on that port. To verify that a port is in the shutdown state, you should issue the show interfaces type number command, where type and number specify the interface you want to show. A port that has been shut down will be reported as administratively down by the show interfaces type number command.

You are not enabling port security on an access port by issuing the commands in this scenario. To protect switch interfaces against Media Access Control (MAC) flooding attacks, you should enable port security on all access mode interfaces on the switch. Issuing the switchport port-security command in interface configuration mode enables port security with default settings. You can modify port security settings before you enable port security by issuing the switchport port-security mac-address mac-address command, the switchport port-security maximum maximum-number-of-mac-addresses command, and the switchport port-security violation [protect | restrict | shutdown] command.

When enabled with its default settings, port security will shut down a port on which a violation occurs. In addition, port security will allow only the first MAC address to connect to the port to access the port.

Day 49 - Port Security

Included Exam B items: Q39

Original Exam B Question 39

You issue the following commands on SwitchA:

```
SwitchA#configure terminal
```

```
SwitchA(config)#interface fastethernet 0/8
```

```
SwitchA(config-if)#switchport port-security
```

No other port security commands have been applied to the FastEthernet 0/8 interface.

Which of the following will occur if more than the maximum number of allowable devices attempt to access the network through the FastEthernet 0/8 interface on SwitchA? (Select the best answer.)

A. The switch will shut down the interface, but no other interfaces will be affected.

The switch will continue to allow traffic through the interface, and no log file will be

B.

generated.

The switch will continue to allow traffic through the interface, and a log file will be

° cC.

generated.

D. . The switch will shut down, and no further traffic will be allowed through the switch.

Correct Answer: A

Explanation

The switch will shut down the interface, but no other interfaces will be affected if more than the maximum number of allowable devices attempt to access the network through the FastEthernet 0/8 interface on SwitchA. In this scenario, the switchport port-security command has been issued to enable port security on the FastEthernet 0/8 interface. The scenario also states that no other port security commands have been applied to the FastEthernet 0/8 interface. Therefore, the following default port security settings are applied:

- Sticky address learning is disabled.
- A maximum of one Media Access Control (MAC) address will be allowed on the port.

The Shutdown violation mode is enabled.

- The port security aging time is configured to zero.
- The port security static aging is disabled.

The port security aging type is configured to absolute.

You can use port security to limit traffic on a switch port so that only traffic sent from one or more statically configured or dynamically learned source MAC addresses is allowed into a switch port.

When port security is enabled on a port, a switch checks the source MAC address of inbound traffic to see whether that MAC address is authorized to send traffic through that switch port. The default action is to place the interface into an err-disabled state, which effectively shuts down the interface; therefore, interface FastEthernet 0/8 will shut down if more than one device attempts to send traffic through the port. This action will not affect any other ports on the switch.

You can issue the switchport port-security violation command to modify the action taken when more than the maximum number of addresses are learned. The switchport port-security violation command can be issued with either the restrict keyword, the protect keyword, or the shutdown keyword. The restrict keyword configures the interface to remain up if more than the maximum number of addresses are learned, but traffic from the violating devices is dropped and a log entry is generated. Similarly, the protect keyword configures the interface to remain up if more than the maximum number of addresses are learned and to drop traffic from violating devices; however, unlike the restrict keyword, the protect keyword does not create a log of the violations. The shutdown keyword configures the port to shut down if more than the maximum allowed MAC addresses are learned on the interface.

Day 51 - Dynamic ARP Inspection

Included Exam B items: Q34

Original Exam B Question 34

You issue the show vlan brief command on Switch1 and receive the following partial output:

```
Switch1#show vlan brief
1 default active Gi0/1, Gi0/2
11 =VLAN0011 active Fa0/1, Fa0/2, Fa0/3, Fa0/4
Fa0/5, Fa0/6, Fa0/7, Fa0/8
Fa0/9, Fa0/10
12 ~VLAN0012 active Fa0/11, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17
Fa0/18
Fa0/19
14 active Fa0/20, Fa0/21, Fa0/22
Fa0/23
Fa0/24
```

You issue the following commands on Switch1:

```
Switch1#configure terminal
Switch1(config)#ip arp inspection vlan 11-12,14
Switch1(config)#interface range gigabitethernet 0/1 - 2
Switch1(config-if-range)#switchport access vlan 1
Switch1(config-if-range)#switchport mode access
```

Which of the following statements is true? (Select the best answer.)

- A. Only GigabitEthernet 0/1 and GigabitEthernet 0/2 ports are untrusted ports.
- B. Only GigabitEthernet 0/1 and GigabitEthernet 0/2 are trusted ports.
- C. DAI is configured on only GigabitEthernet 0/1 and GigabitEthernet 0/2.
- D. DAI is not configured on GigabitEthernet 0/1 and GigabitEthernet 0/2.

Correct Answer: D

Explanation

Dynamic ARP Inspection (DAI) is not configured on GigabitEthernet 0/1 and GigabitEthernet 0/2 in this scenario. The ip arp inspection vlan 11-12,14 command does not include virtual local area network (VLAN) 1. In addition, neither the ip arp inspection command nor the ip arp inspection trust command has been issued on the GigabitEthernet interfaces. Therefore, DAI is not enabled on VLAN 1. The switchport access vlan 1 command that has been issued on both the GigabitEthernet 0/1 interface and the GigabitEthernet 0/2 interface in this scenario configures those ports to reside on VLAN 1 if the ports are operating access mode. The switchport mode access command configures the ports to operate in access mode.

When DAI is configured on a switch, ports are labeled as either trusted or untrusted. An Address Resolution Protocol (ARP) packet that is received on a trusted port is not inspected, whereas an ARP packet received on an untrusted port is inspected. If an ARP reply received on an untrusted port does not match a trusted entry on the switch, the packet will be dropped.

DAI can be enabled on a single VLAN or on multiple VLANs. To enable DAI, you should use the ip arp inspection vlan global configuration command. The syntax of the ip arp inspection vlan command is ip arp inspection vlan {vlan-ID | vlan-range}. A range of VLANs can be entered by using a comma-separated list and/or a dash-separated pair of VLAN numbers indicating the range of VLANs to include. For example, each of the following commands enables DAI on VLANs 2 through 4: ip arp inspection vlan 2,3,4 and ip arp inspection vlan 2-4. Configuring DAI on each VLAN ensures that traffic sent from each host is inspected. In addition, each port is by default an untrusted port.

By default, a port is configured as an untrusted port when DAI is enabled on that port. Therefore, configuring VLANs 11, 12, and 14 with DAI by issuing the ip arp inspection vlan 11-12,14 command ensures that any port operating in those VLANs is automatically an untrusted port.

When DAI is configured for an entire VLAN, you can override the default configuration for a given port by issuing the ip arp inspection trust command in interface configuration mode. In this scenario, however, VLAN 1 has not been globally configured to use DAI.

Day 53 - WAN Architectures

Included Exam B items: Q13, Q27

Original Exam B Question 13

Which of the following digital networking technologies has a total bandwidth of 1.544 Mbps? (Select the best answer.)

- A. E3
- B. T3
- C. E1
- D. T1

Correct Answer: D

Explanation

T1 is a digital networking technology that has a total bandwidth of 1.544 megabits per second (Mbps). To achieve the 1.544-Mbps rate, T1 connections encapsulate data in DS1 signaling frames. DS1 signaling frames include 24 DSO channels and 1 framing bit. Each of the 24 DSO channels transfers 8 bits of data at a time. The inclusion of the framing bit raises the total amount of data in a single DS1 frame to 193 bits. T1 connections transmit 8,000 DS1 frames per second, which brings the total bandwidth to 1,544,000 bits per second (bps), or 1.544 Mbps. T1 connections typically transfer data across two pairs of shielded copper wires, known as a T1 line, which extend from the service provider to the customer. One pair of wires within the T1 line is used to send data, whereas the other pair is used to receive data. T1 is primarily used in the United States and Japan.

E1 is a digital networking technology, but it does not have a total bandwidth of 1.544 Mbps. Instead, E1 connections have a total bandwidth of 2.048 Mbps. Primarily used in Europe, E1 connections operate similarly to T1 connections. However, instead of using 24 eight-bit DSO channels, E1 connections use 32 eight-bit timeslots. E1 connections transmit 8,000 timeslot segments per second, which equals 2,048,000 bits per second (bps), or 2.048 Mbps. Like T1s, E1 connections transfer data across two pairs of shielded copper wires.

T3 is a digital networking technology, but it does not have a total bandwidth of 1.544 Mbps. Instead, T3 connections have a total bandwidth of 44.736 Mbps. Primarily used in the United States and Japan, T3 connections transmit data across 672 DSO channels 8,000 times per second. T3 connections are usually too cost prohibitive for use except by Internet and telephone service providers.

E3 is a digital networking technology, but it does not have a total bandwidth of 1.544 Mbps. Primarily used in Europe, E3 connections have 16 E1 lines and a total bandwidth of 34.368 Mbps. E3 connections are usually too cost prohibitive for use except by Internet and telephone service providers.

Original Exam B Question 27

The sending device on a site-to-site VPN that is constructed by using GRE with IPSec for transport encrypts a new packet.

Which of the following steps occurs next? (Select the best answer.)

- A. The sending device sends the packet to the destination.
- B. The sending device adds the session key to the packet.
- C. The receiving device decrypts the packet.
- D. The sending device adds a VPN header and an IP header.

Correct Answer: D

Explanation

After a sending device on a site-to-site virtual private network (VPN) that is constructed by using Generic Routing Encapsulation (GRE) with Internet Protocol Security (IPSec) for transport encrypts a new packet, the sending device adds a VPN header and an Internet Protocol (IP) header to the packet in a process known as encapsulation.

A site-to-site VPN uses IPSec to transport information across a tunnel that is established between two devices. A typical site-to-site VPN uses GRE with confidentiality, integrity, and antireplay protection provided by IPSec. There are four steps in the site-to-site VPN IPSec encryption process. By contrast, a remote access VPN uses client software to encrypt traffic between a remote user and internal company resources, typically over the Internet.

First, the sending device combines a session key, which is also known as an encryption key or a shared key, with the data that is to be transported over the tunnel. It then uses the session key to encrypt both the data and the key.

Second, the sending device encapsulates the encrypted data and session key into a packet with a VPN header and a new IP header. These headers contain the source and destination information that is used to transport the encrypted data and session key over the tunnel.

Third, the sending device sends the completed packet to the destination device at the other end of the tunnel, or site-to-site VPN.

Fourth and finally, the destination device, or receiving device, uses the same session key that the sending device used for encryption to decrypt the encrypted packet and session key.

Day 54 - Virtualization & Cloud

Included Exam B items: Q50

Original Exam B Question 50

Which of the following statements is true regarding VMs? (Select the best answer.)

VMs running on a host computer must run the same version of an OS as the host

fo)

computer.

B. All software is shared between the host computer and the VMs.

C. Multiple VMs can be running simultaneously on a single host computer.

Installing virus protection on the host computer automatically protects any VMs running on that host computer.

Correct Answer: C

Explanation

Multiple virtual machines (VMs) can be running simultaneously on a single host computer. A VM is an isolated environment running a separate operating system (OS) while sharing hardware resources with a host machine's OS. For example, you can configure a Windows 10 VM that can run within Windows 11, and both OSs can run at the same time. The Windows 10 VM could then be used as a testing environment for patch or application deployment. Depending on a computer's hardware capabilities, multiple VMs can be installed on a single computer, which can help provide more efficient utilization of hardware resources.

Before a VM is installed, it is important to ensure that the hardware on the host in which you are configuring the VM has enough central processing unit (CPU) process availability and random access memory (RAM) to support the simultaneous use of multiple OSs and that the client you are accessing the VM from has sufficient network bandwidth. An additional programming layer, known as a hypervisor, is required in order for the VM to communicate with the host hardware. A hypervisor is used to allocate hardware resources, such as hard drive space, CPU, and RAM, to the VM. Each VM in a hypervisor runs separately from other VMs.

The VMs on a host computer can, but are not required to, run the same version of an OS as the host computer. For example, you can install Ubuntu Linux on a VM that is hosted on a Windows 10 computer. Alternatively, as in the example given previously, you can configure a Windows 10 VM that can run within Windows 8.

Installing virus protection on the host computer will not automatically protect any VMs running on that host computer. Securing the host computer does not secure all virtual computers running on that host computer. You must manually manage the security of each VM that is installed on a host computer. For example, installing patches and security software on the host computer will not also configure the patches and software to be installed on the VMs.

Although a VM shares the hardware resources of the host computer, the software remains separate. Software installed on the host is not accessible from within the VM. For example, Microsoft Office might be installed on the host computer, but in order to access Microsoft Office from within a VM you must also install Microsoft Office on the VM. Separate instances of software on the host computer and on each VM can help protect the host computer from potentially harmful changes made within a VM. For example, if a VM user accidentally deletes a system file or installs malicious software, the host computer will not be affected.

Day 55 - Wireless Fundamentals

Included Exam B items: Q6, Q35

Original Exam B Question 6

Which of the following can be implemented to provide network-assisted power savings in a wireless network? (Select the best answer.)

- A. 802.11W
- B. 802.11r
- C. 802.11k
- D. 802.11v

Correct Answer: D

Explanation

The Institute of Electrical and Electronics Engineers (IEEE) 802.11v standard provides network-assisted power savings in a wireless network. With 802.11v, a wireless client can remain in a standby or sleep state longer to improve battery life. 802.11v provides a Directed Multicast Service (DMS), which enables a client to request that certain multicast messages be delivered as unicast messages instead. This enables the client to ignore the multicast messages during sleep and to receive the unicast messages at a greater rate of speed once it is active again. In addition, 802.11v provides a client with the ability to extend the amount of time it may remain idle before it is disassociated with an access point (AP). This enables the client to wake less frequently from sleep and extends the life of the client's battery.

The IEEE 802.11r standard provides fast transition roaming, not network-assisted power savings, in a wireless network. With 802.11r, a roaming wireless client and an AP perform the Pairwise Master Key (PMK) calculation in advance. These precalculated PMK keys are then used after the client sends its reassociation request to the new AP and the AP responds with its reassociation response. This process enables the client to transition to a new AP without needing to reauthenticate, thereby speeding the handover between APs and minimizing adverse Quality of Service (QoS) effects caused by roaming. The IEEE 802.11w standard provides management frame protection (MFP), not network-assisted power savings, in a wireless network. MFP is a security mechanism that prevents management frames from being compromised by a malicious user. Management frames are used to manage the connection between an AP and a wireless client. Numerous types of management frames exist: beacons, probe requests, probe responses, association requests, association responses, authentication requests, authentication responses, deauthentications, reassociation requests, reassociation responses, and announcement traffic indication messages. Each of these frame types performs specific duties necessary to initialize, maintain, or terminate wireless transmissions between an AP and a client. Because management frames are not encrypted, they are frequently targeted by malicious users who want to disrupt network access via a Denial of Service (DoS) attack. In a DoS attack, an attacker floods a host with packets in an attempt to overload the host so that it cannot respond to further requests.

The IEEE 802.11k standard provides-assisted roaming in a wireless network, not network-assisted power savings, in a wireless network. With 802.11k, a roaming wireless client can request a list of known neighboring APs that might be potential candidates for transition. The list returned by an AP is optimized for the client and contains only APs in the same wireless band with which the client is associated. This enables the client to find a new AP without needing to probe unnecessary wireless channels, which speeds transition and increases battery life for portable devices.

Original Exam B Question 35

One AP is installed on the network you administer. All of the wireless clients on the network have been configured with the SSID of the AP and can connect to the AP.

Which of the following actions would cause all of the wireless clients on the network to lose connectivity? (Select 2 choices.)

- A. changing the SSID on the AP
- B. changing the channel on the AP
- C. changing the encryption key on the AP
- D. disabling SSID broadcasts on the AP

Correct Answer: A, C

Explanation

Changing the encryption key or the Service Set Identifier (SSID) on the access point (AP) would cause all of the wireless clients on the network to lose connectivity. All clients must be configured with the same encryption key and SSID that are configured on the AP.

Disabling SSID broadcasts on the AP would not cause the wireless clients on the network to lose connectivity. When SSID broadcasts are disabled, the SSID is not broadcast in beacon frames on a wireless network. For wireless clients to associate with an AP on which SSID broadcasts are disabled, you must manually configure each wireless client with the hidden SSID. Because all of the wireless clients have already been configured with the SSID of the AP, disabling SSID broadcasts would not cause them to lose connectivity. However, you would need to manually configure new wireless clients with the hidden SSID.

Changing the wireless channel on the AP could cause some wireless clients to receive a stronger signal, a weaker signal, or no signal at all due to interference. However, it is unlikely that changing the channel on the AP would cause all of the wireless clients on the network to lose connectivity. To ensure that the devices on your network do not lose connectivity due to interference or signal blockage, you should install multiple APs on the network. When using multiple APs, you should overlap the signal coverage areas by at least 10 percent and use nonoverlapping channels for each AP. Overlapping channels can cause users to experience slow wireless network performance.

Day 56 - Wireless Architectures

Included Exam B items: Q9, Q20, Q56, Q71

Original Exam B Question 9

Which of the following best describes an AP deployment that connects APs to a WLC that is housed within a switch stack? (Select the best answer.)

- A. autonomous AP deployment
- B. cloud-based AP deployment
- C. embedded AP deployment
- D. lightweight AP deployment

Correct Answer: C

Explanation

Of the available choices, an embedded access point (AP) deployment typically connects APs to a Cisco wireless LAN controller (WLC) that is housed within a switch stack. An AP is a device that connects a wireless client to a wired network. The primary difference between this deployment and others is that the WLC is embedded within a stack of switching hardware instead of existing as a separate entity. APs can connect to the WLC by connecting to switches that are directly hosting the WLC or switch ports that are operating on the same virtual local area network (VLAN) as the WLC. A lightweight AP deployment can be an embedded AP deployment. However, a lightweight AP deployment does not always connect APs to a WLC that is housed within a switch stack. A lightweight AP deployment requires a separate wireless controller. Wireless clients connect to lightweight APs, which are capable of performing real-time wireless network functions but rely on a WLC for management functions. The connection between a lightweight AP and a WLC is created by using two tunnels established by the Control and Provisioning of Wireless Access Points (CAPWAP) tunneling protocol. Information sent between lightweight APs and the WLC is encapsulated in Internet Protocol (IP) packets. This process enables a lightweight AP and the WLC to manage connectivity to the same wireless local area network (WLAN) yet be separated by both physical and logical means. This type of deployment is also known as a split-MAC architecture because the lightweight AP handles the frames while the WLC handles the management functions.

An autonomous AP deployment does not connect APs to a WLC that is housed within a switch stack. An autonomous AP contains network interfaces for both wireless and wired networks; it is typically deployed as part of an autonomous AP architecture in which APs are connected directly to the access layer of the three-tier hierarchical network model.

A cloud-based AP deployment does not connect APs to a WLC that is housed within a switch stack. Instead, cloud-based APs connect to and are automatically configured by a WLC that is housed in a cloud-based system. For example, a Cisco Meraki AP provides wireless access by connecting to a centralized management system known as the Cisco Meraki Cloud. APs deployed at the access layer of the three-tier hierarchical network model contact the cloud in order to automatically configure themselves. APs are managed through a cloud-based dashboard.

Original Exam B Question 20

Which of the following WLC interfaces is typically used for continuous SNMP polling? (Select the best answer.)

- A. the management interface
- B. the AP-manager interface
- C. the service port interface
- D. interface
- E. the virtual interface

Correct Answer: A

Explanation

The management interface on a wireless LAN controller (WLC) is typically used for continuous Simple Network Management Protocol (SNMP) polling. A WLC can contain up to four static interfaces: the management interface, the AP-manager interface, the virtual interface, and the service port interface. The management interface on a WLC is used for in-band management information. This interface is

used for all Layer 2 LWAPP communications between the controller and the lightweight access points (APs) and to communicate with other WLCs on the wireless network.

The AP-manager on a WLC controls all Layer 3 communications between a WLC and a lightweight AP. The AP-manager interface contains the Internet Protocol (IP) address that is used as the source IP address by which the lightweight APs communicate with the WLC. Because the AP-manager interface communicates with the lightweight APs on the wireless network, the IP address assigned to the AP-manager interface should be unique on the network. After the interface has been configured, the WLC uses the AP-manager interface to listen for Layer 3 Lightweight Access Point Protocol (LWAPP) communications.

The service port interface is used for out-of-band maintenance purposes on a WLC. This interface is a physical interface on the WLC that can be used to recover the WLC in the event that the WLC fails. The service port interface can be accessed by using Telnet or Secure Shell (SSH) and is the only interface that is available while the WLC is booting.

The virtual interface can be used for mobility management by providing a specific IP address that is the same across multiple controllers when wireless clients roam among the controllers. This enables seamless roaming among the controllers. The virtual interface is also used for embedded Layer 3 security where web authorization has been enabled for clients; the user is redirected to the IP address of the virtual interface when the user opens a web browser. In addition, if Dynamic Host Configuration Protocol (DHCP) relay has been enabled on the controller, the virtual interface can be used as the DHCP server address on wireless clients.

In addition to the four static interfaces, a WLC can contain up to 512 dynamic interfaces. Dynamic interfaces are user-defined and are typically used for wireless client data. The dynamic interfaces function similarly to virtual local area networks (VLANs). For example, you can create a dynamic interface to segment traffic on the WLC.

Original Exam B Question 56

Which of the following Cisco lightweight AP modes do not provide client connectivity? (Select 3 choices.)

- A. local mode
- B. rogue detector mode
- C. FlexConnect mode
- D. bridge mode
- E. monitor mode
- F. sniffer mode

Correct Answer: B, E, F

Explanation

Lightweight access point (AP) monitor mode, sniffer mode, and rogue detector mode do not provide client connectivity. After adding a lightweight AP to a wireless LAN controller (WLC), you can configure the lightweight AP to operate in one of six modes depending on your needs and the capability of the lightweight AP. Three of the modes, monitor mode, sniffer mode, and rogue detector mode, are used primarily for monitoring and administrative purposes and, thus, do not provide client connectivity. For example, you can place a lightweight AP in monitor mode to configure the lightweight AP to scan wireless traffic in search of rogue lightweight APs. You can place a lightweight AP in sniffer mode to configure the lightweight AP to scan for wireless traffic and to send the traffic to a management station running a tool such as Wireshark. You can configure a lightweight AP to operate in rogue detector mode to configure the lightweight AP to scan traffic on the wired connection in search of rogue lightweight APs and rogue clients on the wired network.

FlexConnect mode allows for client connectivity. A lightweight AP operating in FlexConnect mode can be remotely managed via a wide area network (WAN) link, which is useful when lightweight APs are deployed in remote offices. This enables administrators to deploy a lightweight AP in the remote office without also needing to deploy a WLC to the office. The lightweight AP can thereby provide wireless connectivity to clients without sending the data to the WLC. Furthermore, lightweight APs operating in FlexConnect mode can provide client connectivity even if the connection to the remote WLC is lost.

A lightweight AP is in local mode by default. In this mode, the lightweight AP can connect to a WLC and can provide client connectivity. In addition, a lightweight AP operating in local mode monitors all wireless channels.

A lightweight AP operating in bridge mode can provide client connectivity. You can configure a

lightweight AP for bridge mode in order to use the lightweight AP to connect two separate network segments. In addition, you can use this mode to configure a mesh network.

Original Exam B Question 71

Which of the following statements are true regarding dynamic interfaces on WLCs? (Select 2 choices.)

- A. Dynamic interfaces are typically used for client data.
- B. Dynamic interfaces are typically used for management information.
- C. Dynamic interfaces must be reachable by other WLCs.
- D. Dynamic interfaces are user-defined.
- E. Dynamic interfaces are often used for maintenance purposes.

Correct Answer: A, D

Explanation

Dynamic interfaces are user-defined and are typically used for client data. A wireless LAN controller (WLC) contains both static and dynamic interfaces. A WLC can contain up to four types of static interfaces: the management interface, the AP-manager interface, a virtual interface, and the service port interface. A WLC can contain up to 512 dynamic interfaces. The dynamic interfaces function similarly to virtual local area networks (VLANs). For example, you can create a dynamic interface to segment traffic on the WLC.

The management interface, which is a static interface, is used for management information. This interface is used for all Layer 2 Lightweight Access Point Protocol (LWAPP) communications between the controller and the lightweight access points (APs). In addition, the management interface is used to communicate with other WLCs on the wireless network.

The service port interface, which is a static interface, is used for out-of-band maintenance purposes on a WLC. This interface is a physical interface on the WLC that can be used to recover the WLC in the event that the WLC fails. The service port interface is the only interface that is available while the WLC is booting. The service port can be used as a last-resort measure to access the WLC graphical user interface (GUI).

Alternatively, you can use the console port to access the WLC command-line interface (CLI). The console port is an out-of-band asynchronous interface and is typically accessed with a serial cable or a universal serial bus (USB) to RJ-45 adapter.

It is not necessary for a dynamic interface to be reachable by all other WLCs. The WLCs will use the management interface, not a dynamic interface, to exchange information.

Day 57 - Wireless Security

Included Exam B items: Q8, Q67

Original Exam B Question 8

You are configuring Layer 2 security on a WLAN by using the WLC GUI. You select WPA+WPA2 from the Layer 2 Security drop-down list box. You then select the MAC Filtering option to ensure that only specific clients can join, and you select the WPA2 Policy option to ensure that WPA2 is used. You want to configure the WPA2 passphrase in ASCII format.

Which of the following WPA2 key management methods should you select from the Auth Key Mgmt drop-down list box? (Select the best answer.)

- A. 802.1X
- B. CCKM
- C. 802.1X+CCKM
- D. PSK

Correct Answer: D

Explanation

You should select the PSK Wi-Fi Protected Access 2 (WPA2) key management method from the Auth Key Mgmt drop-down list box if you want to configure the WPA2 passphrase in American Standard Code for Information Interchange (ASCII) format. The PSK method configures WPA or WPA2 to use the Pre-Shared Key (PSK) key management method. This method requires that an administrator configure each wireless client that will connect to the network with the key that is configured on the Cisco wireless LAN controller (WLC). The PSK option supports key entry as either an ASCII passphrase from 8 through 63 characters in length or a key of 64 hexadecimal values. Combining WPA or WPA2 with a PSK key management method is often known as WPA-PSK, or WPA Personal.

You should not select the 802.1X key management method in this scenario. The Institute of Electrical and Electronics Engineers (IEEE) 802.1X standard defines a method of port-based network access control. On Cisco wireless local area networks (WLANs), the 802.1X key management method is the default method for both WPA and WPA2. It typically requires a Remote Authentication Dial-In User Service (RADIUS) server and uses various Extensible Authentication Protocol (EAP) implementations to authenticate users. Combining WPA or WPA2 with an 802.1X key management method is often known as WPA-8021X mode, or WPA Enterprise.

You should not select the CCKM key management method in this scenario. This option enables the Cisco Centralized Key Management (CCKM) key management method. CCKM is a Cisco-proprietary fast-rekeying method that enables a wireless client to roam from one access point to another without requiring intervention from the WLC. CCKM is typically used to reduce delay when wireless clients transition between access points so that delay-sensitive services, such as Voice over Internet Protocol (VoIP), operate smoothly.

You should not select the 802.1X+CCKM key management method in this scenario. This option enables 802.1X clients to use the CCKM key management method to roam between access points without performing the complete authentication process again. Normally, 802.1X clients mutually authenticate to a new access point. This process likewise involves reauthenticating with the RADIUS server. The 802.1X+CCKM key management method removes the need to reauthenticate with the RADIUS server, thus reducing the amount of time it takes for an 802.1X client to roam between access points.

Original Exam B Question 67

Which of the following are used by WPA2 to provide MICs and encryption? (Select 2 choices.)

- A. TKIP
- B. GCMP
- C. RC4
- D. AES
- E. CCMP

Correct Answer: D, E

Explanation

Advanced Encryption Standard (AES) and Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP) are used by Wi-Fi Protected Access 2 (WPA2) to provide message integrity checks (MICs) and encryption. Wireless security protocols use MICs to prevent data tampering. Encryption is used to protect confidentiality.

WPA2, which implements the 802.11i wireless standard, was developed to address the security vulnerabilities in the original WPA standard. One enhancement over WPA included in WPA2 is the encryption algorithm. AES is a stronger encryption algorithm than the RC4 algorithm used by earlier wireless standards. When AES is implemented, a 128-bit block cipher is used to encrypt data and a security key of 128, 192, or 256 bits can be used. This is a processor-intensive operation, and implementing WPA2 and AES often requires new hardware, such as new wireless access points (APs) and new client wireless network adapters.

In addition to AES, WPA2 also uses CCMP to provide encryption. CCMP is an encryption mechanism that uses block ciphers. In WPA2, CCMP is used by AES during the encryption process. The WPA2 encryption process is thus sometimes known as AES-CCMP.

RC4 is a stream cipher encryption algorithm used in the Wired Equivalent Privacy (WEP) protocol. Unlike AES, which supports an encryption key length of 256 bits, RC4 supports an encryption key length of up to 128 bits. Consequently, RC4 is not as secure as AES. Furthermore, RC4 uses a stream cipher, which is a less secure encryption method. RC4 is not used with WPA2.

Temporal Key Integrity Protocol (TKIP) is used to provide MICs and encryption in the WPA protocol. WPA is the successor to WEP and the predecessor of WPA2. The WPA TKIP implementation provides improvements over WEP but uses RC4 as the encryption algorithm. TKIP supports an encryption key of up to 128 bits, whereas AES supports an encryption key of 256 bits. Consequently, TKIP is not specified as the encryption method in the 802.11i standard.

Galois/Counter Mode Protocol (GCMP) is used along with AES to provide MICs and encryption in the WPA3 protocol. The WPA3 protocol was introduced in 2018 as a future replacement for WPA2. GCMP is considered to be stronger and more efficient than CCMP. GCMP uses AES to provide encryption and Galois Message Authentication Code (GMAC) to provide MICs. In addition, WPA3 uses Simultaneous Authentication of Equals (SAE) and introduces support for Protected Management Frames (PMF). SAE replaces the use of pre-shared keys (PSKs) and enables clients and access points (APs) to mutually authenticate. PMF secures management frame communication between a wireless AP and a client.

Day 58 - Wireless Configuration

Included Exam B items: Q57

Original Exam B Question 57

You want to use an HTTPS connection to access and configure a WLC.

Which of the following commands should you issue? (Select the best answer.)

- A. config network webmode enable
- B. config webmode https enable
- C. config network https enable
- D. config network secureweb enable

Correct Answer: D

Explanation

You should issue the config network secureweb enable command to configure a wireless LAN controller (WLC) to support Secure Hypertext Transfer Protocol (HTTPS) management connections. Because secure mode is enabled by default, you normally do not need to issue this command unless you have previously disabled secure access to the WLC and you want to re-enable secure access to the WLC.

You can use a command-line interface (CLI) or a graphical user interface (GUI) to configure a WLC. To access the CLI of a WLC, you can either establish a local serial connection to the console port of the WLC or establish a remote console session over an Ethernet connection. To access the management GUI of a WLC, you should type https://ip-address in a browser window, where ip-address is the Internet Protocol (IP) address of the WLC.

By default, HTTP management access is disabled on a WLC; however, you can issue the config network webmode enable command to enable HTTP management connections. Likewise, to disable HTTP management access, you can issue the config network webmode disable command.

The config webmode https enable and config network https enable commands use incorrect syntax and are not valid Cisco commands.

Day 60 - JSON, XML & YAML

Included Exam B items: Q33

Original Exam B Question 33

A REST API query returns the following output:

```
{
  "id": 12345,
  "fname": "John",
  "lname": "Doe",
  "group": {
    "role": "Receivables",
    "read-only": ["Accounting Folder", "Sales Folder"]
  }
}
```

Which of the following statements is true? (Select the best answer.)

- A. The value of the fname key is equal to the lname key and its value.
- B. The value of the lname key is a text value.
- C. The value of the group key is an array.
- D. The value of the id key is an array.
- E. The value of the read-only key is a text value.
- F. The value of the role key is an object.

Correct Answer: B

Explanation

Of the available choices, it is true that the value of the lname key in the Representational State Transfer (REST) Application Programming Interface (API) output above is a text value. REST is an API architecture that uses Hypertext Transfer Protocol (HTTP) or HTTP Secure (HTTPS) to enable external resources to access and make use of programmatic methods that are exposed by the API. The REST API that is being queried in this scenario has returned output in JavaScript Object Notation (JSON) format. JSON is a data modeling language that is commonly used by REST APIs. The JSON data modeling language returns data in the form of an object that contains key and value pairs.

A single JSON object can contain multiple key and value pairs. Each key and value pair inside a JSON object is separated from the others by a comma (.). Furthermore, each pair's key is separated from its value by a colon (:). The element in quotation marks on the left side of each colon is the key. The element on the right side of each colon is the value, which might or might not be enclosed in quotation marks. There are several data value types that can be returned in JSON output: text, numeric, array, object, Boolean, and null.

The value of the lname key, the value of the fname key, and the value of the role key are all text values in the JSON object in this scenario. JSON keys and JSON text values are always enclosed in quotation marks. Text values typically contain alphanumeric or Unicode characters.

The value of the id key is numeric in the JSON object in this scenario. Numeric data is defined by digits that are not enclosed in quotation marks. If a string of digits is enclosed in quotation marks, it will be treated as text even if it contains only numeric characters. The numeric value of the id key in this scenario is 12345.

The value of the group key in this scenario is another JSON object. A JSON object can be identified by the curly brackets that mark the beginning and the end of the object. A left curly bracket ({) marks the beginning of a JSON object. A right curly bracket (}) marks the end of a JSON object. JSON objects can be nested as values inside other JSON objects. In addition, JSON objects can be specified as values inside other JSON objects. In this scenario, the value of the group key in the primary JSON object is shown in the following JSON object:

```
role": "Receivables",
read-only ": [
Accounting Folder ",
Sales Folder ]
```

As previously mentioned, a JSON object is a group of key and value pairs. The JSON object above contains two key and value pairs. The role key contains a text value of Receivables. The read-only

key, on the other hand, contains an array.

The value of the read-only key in this scenario is an array that contains two text values: Accounting Folder and Sales Folder. Unlike JSON objects, arrays contain values only. They do not contain key and value pairs. Arrays are typically contained within square brackets. A left square bracket ([) indicates the beginning of the array. A right square bracket (]) indicates the end of the array. Each value inside an array is separated from the others by a comma (,). Although the array in this scenario contains two text values enclosed in quotation marks, an array can contain any of the other JSON data types. For example, an array can include JSON objects or other arrays as values. In addition, a given array need not contain only a single data type.

There are no Boolean or null values in the JSON output in this scenario. A given JSON key can be assigned a Boolean value of either true or false. Because these are Boolean values, not text values, they are typically not placed inside quotation marks. Null values technically have no value at all, although it is not exactly the same as a numeric value of 0. A given JSON key can be assigned a null value by specifying null without quotation marks.

Day 61 - REST APIs

Included Exam B items: Q38, Q74

Original Exam B Question 38

Which of the following statements about REST APIs are true? (Select 2 choices.)

- A. REST APIs are typically used to communicate with an SDN application plane.
- B. REST APIs are typically used to communicate with an SDN data plane.
- C. REST APIs encode data exclusively in XML format.
- D. REST APIs encode data exclusively in JSON format.
- E. REST APIs encode data in either XML format or JSON format.

Correct Answer: A, E

Explanation

Representational State Transfer (REST) Application Programming Interfaces (APIs) encode data in either Extensible Markup Language (XML) format or in JavaScript Object Notation (JSON) format. In addition, REST APIs are typically used to communicate with a Software-Defined Networking (SDN) application plane.

An SDN controller uses two different sets of APIs: one set to communicate with applications and another set to communicate with devices. Northbound APIs enable an SDN controller to communicate with applications in the application plane. Applications use northbound APIs to send requests or instructions to the SDN controller, which uses that information to modify and manage network flow. Southbound APIs enable an SDN controller to communicate with devices in the data plane.

XML is a markup language that is similar to Hypertext Markup Language (HTML) in structure; it uses tags to define blocks of data. Whereas HTML is used to render information on a webpage, XML is a more structured language that is used to format data in a way that can be easily transmitted over the Internet and parsed by a variety of applications.

JSON is a data modeling language that returns data in the form of an object that contains key and value pairs. A single JSON object can contain multiple key and value pairs. Each key and value pair inside a JSON object is separated from the others by a comma (.). Furthermore, each pair's key is separated from its value by a colon (:). The element in quotation marks on the left side of each colon is the key. The element on the right side of each colon is the value, which might or might not be enclosed in quotation marks. There are several data value types that can be returned in JSON output: text, numeric, array, object, Boolean, and null.

Original Exam B Question 74

A REST API returns an HTTP status code of 301.

Which of the following is the most likely reason this code was returned? (Select the best answer.)

- A. The requested resource has been temporarily moved to another URL.
- B. The requested action was successfully performed on the server.
- C. The requested resource cannot be found.
- D. The requested resource has been permanently moved to another URL.

Correct Answer: D

Explanation

Most likely, the requested resource has been permanently moved to another Uniform Resource Locator (URL) if a Representational State Transfer (REST) Application Programming Interface (API) returns a Hypertext Transfer Protocol (HTTP) status code of 301. HTTP status codes are used to provide information to a client when the client requests information from an HTTP server. In addition, HTTP status codes are returned to API clients that communicate with an API by using HTTP, such as a REST API. When the HTTP 301 status code is returned by a server, the new URL is included in the response.

An HTTP status code of 302, not 301, indicates that the requested resource has been temporarily moved to another URL. The HTTP status code of 302 is typically sent when a client submits a request by using a given URL that is temporarily no longer valid and the server is configured to redirect the client to a temporarily valid URL. This code indicates that the client should continue using the original

URL to make these requests, even though the URL is temporarily invalid.

An HTTP status code of 200, not 301, indicates that the requested action was successfully performed on the server. From a web server, the HTTP 200 code is typically followed by headers and webpage content. From a REST API, the HTTP 200 code is typically followed by headers and output in either JavaScript Object Notation (JSON)-encoded or Extensible Markup Language (XML)-encoded format. An HTTP status code of 404, not 301, indicates that the requested resource cannot be found. This can occur either because the URL is not recognized, because the resource does not exist, or because the client is unauthorized. Typically, an unauthorized client will receive an HTTP 403 status code; however, if the server is configured to hide the existence of a resource from unauthorized clients, the server can instead display an HTTP 404 status code.

Day 62 - Software-Defined Networking

Included Exam B items: Q21, Q68

Original Exam B Question 21

Which of the following Cisco SDA components are typically located north of the DNA Center controller? (Select the best answer.)

- A. the fabric
- B. the applications
- C. the overlay network
- D. the underlay network

Correct Answer: B

Explanation

Of the available choices, applications are the Cisco Software-Defined Access (SDA) components that are typically located north of the Cisco Digital Network Architecture (DNA) Center controller. Cisco DNA Center, which has been rebranded as Cisco Catalyst Center, abstracts and simplifies the complexity of network configuration by implementing a centralized controller, Application Programming Interfaces (APIs), and a graphical user interface (GUI) to automate the configuration of network devices. The Representational State Transfer (REST) API is used to natively communicate with Cisco devices.

To communicate with third-party devices, Cisco DNA Center relies on software development kits (SDKs). The Cisco DNA controller, which is similar to a Software-Defined Networking (SDN) controller, is the central component of a Cisco SDA network. Cisco SDA is a Cisco-developed means of building local area networks (LANs) by using policies and automation.

An SDN controller uses two different sets of APIs: one set to communicate with applications and another set to communicate with devices in the data plane. Northbound APIs enable an SDN controller to communicate with applications in the application plane. Applications use northbound APIs to send requests or instructions to the SDN controller, which uses that information to modify and manage network flow. Southbound APIs enable an SDN controller to communicate with devices in the data plane.

In both SDA and SDN deployments, the controller communicates with devices by using a southbound API. Communication with applications and user interfaces is accomplished by using a northbound API. Of the available choices, the fabric, overlay network, and underlay network are the Cisco SDA components that are typically located south of the Cisco DNA Center controller. The fabric is the entirety of the overlay network and the underlay network.

The overlay network creates Virtual Extensible LAN (VXLAN) tunnels between Cisco SDA switches. The tunnels send and receive traffic between fabric endpoints. The underlay network, on the other hand, is a more traditional network configuration of switches. It is a collection of devices, interfaces, and media that comprises the Internet Protocol (IP) network that connects each fabric node. The underlay network is part of a dynamic discovery process that is involved in creating the overlay network's VXLAN tunnels. When an endpoint in a Cisco SDA network sends traffic to another endpoint, the traffic flows from the endpoint through the overlay network's VXLAN tunnels.

Original Exam B Question 68

To which of the following planes does a centralized controller connect by using a northbound API? (Select the best answer.)

- A. the control plane
- B. the data plane
- C. the application plane
- D. the management plane

Correct Answer: C

Explanation

In a controller-based network, a centralized controller connects to the application plane by using a northbound Application Programming Interface (API). The application plane is the component of a controller-based network in which applications that are written to allow interaction with the centralized controller reside. These applications are typically designed to improve network

management efficiency through network automation. A controller communicates with applications in the application plane by using a northbound API such as Representational State Transfer (REST) or Java Open Services Gateway initiative (OSGi).

In a controller-based network, such as a Software-Defined Networking (SDN) network, the control plane is centralized. The control plane is responsible for network decision making in both a controller-based network and a traditional network. However, the control plane in a traditional network is typically distributed among many devices. The Open Shortest Path First (OSPF) routing protocol running on a series of routers on a traditional network is one example of a traditional control plane. OSPF makes routing decisions for packets that require routing among Layer 3 devices. In a controller-based network, the decision-making logic for packet-handling policies is either moved to a central controller or monitored by a central controller.

In a controller-based network, a centralized controller connects to the data plane by using a southbound API, such as NETCONF, OpenFlow, OpFlex, or OnePK. Layer 2 switches, Layer 3 switches, and end devices typically operate in the data plane. Network tasks that are typically performed in the data plane include the encapsulation and decapsulation of packets, the adding or removing of trunk headers, the matching of Media Access Control (MAC) addresses to a MAC address table, the matching of Internet Protocol (IP) addresses to paths in a routing table, the encryption of data, Network Address Translation (NAT), and filtering by using either access control lists (ACLs) or port security.

In both a controller-based network and a traditional network, the management plane consists of network management protocols, such as Telnet, Secure Shell (SSH), Simple Network Management Protocol (SNMP), and Syslog. All of these protocols enable an administrator to connect to and manage a network device.

Day 63 - Ansible, Puppet, Chef & Terraform

Included Exam B items: Q32

Original Exam B Question 32

What type of file does Ansible use to specify client device host names and other device information? (Select the best answer)

- A. aconfiguration file
- B.
- C. aresource file
- D. atemplate
- O · amodule
- F. aninventory

Correct Answer: F

Explanation

Ansible uses an inventory to specify client device host names and other device information. Ansible is a configuration management tool that operates on macOS, on Linux, or within a Linux virtual machine (VM) on Windows. Unlike other configuration management software packages, such as Terraform, Ansible does not use agent software on managed nodes. Instead, Ansible uses Secure Shell (SSH) to communicate with clients. Clients that are managed by Ansible are agentless, which means that no code needs to be preinstalled on the client device.

An Ansible playbook is used to specify the actions that will be performed on client devices. Playbooks are written in YAML Ain't Markup Language (YAML).

An Ansible template is a file that is written in Jinja2 and contains the configuration that will be used on the client. However, because each client is configured differently, the template contains defined variables. For example, the following template might be used to configure a host device:

```
hostname {{host}}
!
interface fastethernet1/1
ip address {{ipaddr1}} {{submask1}}
!
interface fastethernet1/2
ip address {{ipaddr2}} {{submask2}}
```

Ansible uses a variables file that will substitute values into the template variable definitions. The following is a sample Ansible variable file that is written in YAML:

```
host: RouterA
ipaddr1: 192.168.1.1
submask1: 255.255.255.0
ipaddr2: 192.168.2.1
submask2: 255.255.255.0
```

Ansible does not use configuration files, resource files, or modules; these files are used by Terraform. Terraform is an infrastructure as code (IaC) configuration management tool that can be used to manage cloud-based and local devices. A Terraform configuration file is used to configure the devices. A Terraform resource file specifies the devices, networks, or data stores that you want to manage. Terraform modules are used to organize these configuration and resource files.